USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION

LUKE CAMPBELL MACLEOD
220020677

University of
St Andrews

Project Supervisors: Dr Abd Alsattar Ardatti & Prof Will Lamb

A CS4099 Senior Honours project submitted in partial fulfilment of
the requirements for the degree of:

BSC (HONS) COMPUTER SCIENCE

THE UNIVERSITY OF ST ANDREWS
School of Computer Science

DATE: 03 April 2026

DECLARATION

“ I hereby certify that this dissertation, which is approximately **x** words in length, has been composed by me, that it is the record of work carried out by me and that it has not been submitted in any previous application for a degree. This project was conducted by me at the University of St Andrews from 01/09/2025 to **x** towards fulfilment of the requirements of the University of St Andrews for the degree of BSc Computer Science under the supervision of Dr Abd Alsattar Ardatti and Dr Will Lamb
In submitting this project report to the University of St Andrews, I give permission for it to be published online. I retain the copyright in this work.”

[insert date] [insert signature]

ACKNOWLEDGEMENTS

- Dr. Abd Alsattar Ardatti (The University of St Andrews)
- Prof. Will Lamb (The University of Edinburgh)
- 羅銳邦副教授 - Associate Prof. Ruibang Luo (The University of Hong Kong)
- 楊卓裕博士, Leo - Dr. Leo Yang (The University of Hong Kong)
- Music Contributions: Alice Macmillan, Seoras Chlad, Simeon Fletcher, Cara Morrison, Kate Macmillan, Jack Wolf, Faye Smith
- Bun-Sgoil Steòrnabhaigh (Stornoway Primary School) staff and pupils
- Comhairle Nan Eilean Siar (Western Isles Council) and Aoghnas MacIllinnein OBE (Gaelic Policy Officer) and Dr. Dòmhnall Weir (Gaelic Language Officer)

USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION

ABSTRACT

TBC - Outline of the project using at most 250 words. summarising the research question, methods, results, and conclusions.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF DEFINITIONS

LIST OF ABBREVIATIONS

(tbd)

GME: Gaelic Medium Education

HCI: Human Computer Interaction

GDD: Game Design Document

FR - Functional Requirement

NFR - Non-Functional Requirement

MDA - Mechanics-Dynamics-Aesthetics (game design framework)

WCAG - Web Content Accessibility Guidelines

SVG - Scalable Vector Graphics

UI - User Interface

NPC - Non-Player Character

PII - Personally Identifiable Information

Chapter 1. Introduction

Chan e sinne a nì an cànan, 's e an cànan a nì sinne *Iain Mac a' Ghobhainn*	It is not we who make the language, it is the language that makes us. *Iain Crichton Smith*

The classroom and the home are being reshaped by digital transformation. When technology weaves itself into the fabric of daily life, does one pause to ask what is being lost? Are our voices, language, and culture being discreetly omitted in the process? Or might these perceived digital threats be the very tools that are the key to reclaiming and revitalising what defines us most deeply?

By mid-2026, all 700,000 school-aged pupils in Scotland should have access to a digital device to support their learning, with technology seen as a vital aspect of an education system in the digital age. (The Scottish Government: *Devices for 700,000 children*, 2021). This policy strongly signals that computer interaction for children is no longer a pilot initiative; it is routine. While advancements in digital tools can be beneficial, the design of mainstream technologies may not cater to all users, particularly those in Gaelic Medium Education (GME) who have a unique set of sociolinguistic and pedagogical requirements. Without empirical investigation to formally address these requirements within Human Computer Interaction (HCI), we cannot determine whether current digital resources exclude them or adequately serve their accessibility, interaction, and cultural needs.

Although GME is vital to language revitalisation, its efforts face significant challenges. The way in which GME students interact with the Gaelic language casually is varied, and the introduction of purpose-built technologies could function as a scalable proxy for closing this acquisition-enculturation gap. This project directly addresses these challenges, exploring whether a game built on cultural and linguistic needs can prove effective as a way for upper primary GME students to engage with their language and culture digitally, in a normalised way.

1.1 Problem Statement

Empirical evidence shows a decline in idiomatic Scottish Gaelic among young students. There currently exists no culturally grounded, purpose-built attempt at a game that supports the digital normalisation of Scottish Gaelic for GME students. This project represents a pioneering endeavour to achieve this, using it as a vehicle to transmit idiomatic language.

1.2 Project Roadmap and Methods Overview

This Senior Honours project followed a structured research methodology divided into eight stages: (1) Literature Review, (2) Software Engineering Process, (3) Game Design, (4) Implementation, (5) Ethics, (6) Data Collection, (7) Evaluation and Conclusions, and (8) Future Adaptations.

Stage 1 - Literature Review

To develop a game that genuinely serves those in the GME community, this section integrates three areas of focus:

Table 1 - Literature Review Stages

Part 1	Investigating wider user population, their educational and environmental contexts to ground design decisions that then reflect a real target user group.
Part 2	Drawing on cross-cultural HCI practices to ensure culturally authentic and appropriate representation that goes beyond surface-level localisation.
Part 3	Investigating game design frameworks to identify effective mechanisms for embedding language and cultural content within engaging interactive experiences.

Stage 2 - Software Engineering Process

A hybrid software engineering methodology, water-scrum-fall, is used. There is a linear timeline to this project whereby tasks must be completed in the correct order; however, the implementation of the game itself will adopt an agile scrum approach.

Stage 3 - Design

This project adopts a User-Centred-Design approach to understand the user group's language, culture, and surrounding environment. UCD will be both culturally and academically informed. The game's design motivations will be opened up through personas and a reflexive design process, as well as requirement generation from stakeholders. From there, a Game Design Document will be created.

Stage 4 - Implementation

This stage is where the project materialised, building a functional artifact for use. The game was built as a web application, hosted on Firebase. Project tracking is shown through the Scrum boards in each sprint.

Stage 5 - Ethics

There were 3 phases of significance to this project's ethics, with each stage having sequential requirements from various parties. The first phase involved creating the ethics application, documentation for participants, and, after receiving the URTEC's favourable opinion, recruitment and obtaining legal consent. The second phase was the research session itself, ensuring the activities carried out by participants satisfied all ethical criteria, including the active monitoring of assent and suitable debriefing. The third phase managed the withdrawal window, post-experiment data anonymisation, and governance.

Stage 6 - Data Collection

The final stage of this Senior Honours project was evaluating the artefact. By testing the game in action with the real user group, the research questions set could be answered effectively, as well as uncovering potential areas of improvement. Conducting an on-site experiment with the user group presented an opportunity to gather rich data regarding participant behaviour and needs. To evaluate the game, X pupils carried out the live activity, then filled out a digital Qualtrics questionnaire. Thereafter a discussion group was help to uncover further X.

Stage 7 - Evaluation & Conclusions

TBC

Stage 8 Looking Ahead - Adaptations for Future & Applicability to Other Minority Languages.

TBC (Tertiary Objective)

1.3 Objectives

Below is a list of original objectives of the project as documented in the Description, Objectives, Ethics & Resources (DOER) document.

Primary:

• Conduct a structured literature review on Gaelic and its Education, HCI for minority languages, cross-cultural interaction design, and, focusing on mechanisms for authentic language use in digital platforms.

• Identify and formalise design requirements for a Gaelic digital game, emphasising interface elements and interaction strategies that incentivise active Gaelic usage.

• Prototype a functional digital game that maximises opportunities for users to interact with Gaelic digitally.

Secondary:

• Improve the prototype through teacher and pupil feedback, focusing on usability, cultural engagement, and the effectiveness of Gaelic language integration.

• Map and critically analyse game-based HCI principles against established curricular frameworks, identifying which design choices increase active Gaelic language engagement.
- As this is not a 'serious game', the mapping of HCI principles against curricular frameworks was replaced with game mechanics to carry idiomatic Gaelic was chosen

• Design in-game support systems (tutorials & feedback) that enable self-guided play, minimising the need for adult/supervisor support.

Tertiary:

• Document required adaptations for implementation to future applications.

• Investigate applicability to other minority languages.

• Explore teacher-facing configurations (admin interface)

1.4 Contributions

(To be completed at end of project)

Chapter 2. Literature Review

S e a' Ghàidhlig a th' againn an-diugh, glè thric, mar chuimhneachan air a' Ghàidhlig. Chan e a' Ghàidhlig bheò. Tha i mar bhodhaig air an deach a h-anam a threigsinn. *Aonghas Pàdraig Caimbeul*	The Gaelic we have today is often like a memory of Gaelic. It is not the living Gaelic. It is like a body whose soul has departed. *Angus Peter Campbell*

This chapter introduces the contextual motivation behind the project, explaining some of the barriers affecting this community and why digital artifacts could be of benefit. Below is a summary of each section of this literature review:

Gaelic - A Language Between Tradition and Technology

This section shows a personal narrative of the Gaelic language, then tracks its journey from marginalisation to contemporary revitalisation. It examines challenges facing the biggest revitalisation effort, GME, and identifies a significant gap: the lack of culturally responsive games designed for Gaelic speakers, a problem the community has largely accepted rather than solved.

Cross-Cultural Digital Design

This section examines design principles and methodologies from parallel minority language contexts to establish evidence-based frameworks applicable to the artifact being developed, drawing on research from cultural authenticity (indexical vs. iconic elements), and User-Centred Design approaches. It addresses key design challenges, including balancing cultural authenticity with engagement, integrating culturally normative situations that reflect genuine practices, and ensuring ethical co-design with communities throughout development. The section synthesises global case studies, including indigenous game design principles and successful minority language learning apps, to demonstrate how formal UCD methodologies, respectful cultural representation, and community-led design can create games that transmit both language and cultural identity effectively.

Game Design

This section establishes the theoretical and practical foundations for effective game design by firstly examining the MDA framework as a structured approach to aligning gameplay systems with desired player experiences, and exploring how design decisions around narrative structure, challenge progression, and gamification elements influence engagement and learning outcomes.

2.1 Gaelic - A Language Between Tradition and Technology

Growing up on the Outer Hebrides, my early years were moulded by the language I spoke. Gaelic was the key that opened the cultural doors to my identity. Through that language, I sang songs, learnt instruments, gained an understanding of my ancestors, the meaning behind the names of the land I lived on, and the stories of people who came before me.

Most of my Gaelic was learnt at school through GME, an opportunity unavailable to my parents, who, as a result, never reached fluency in the language. However, it was my grandparents and the broader community who passed down idiomatic Gaelic to me in its most authentic form, outside the boundaries of the classroom.

This personal experience aligns with the conclusions drawn from (Fishman, 1991), who recognised that successful language revitalisation is crucially dependent upon "intergenerational and demographically concentrated home-family-neighbourhood" transmission (Fishman, 1991, p. 395)

2.1.1 Tradition in Transmission: Surviving the Digital Shift

In contemporary Scotland, many GME pupils come from non-Gaelic-speaking households, making the informal experiences described by Fishman unreachable outside of school. Like any other community that is geographically distant or socio-linguistically challenged, technology surfaces as a credible tool to help address these challenges. This raises a critical question: Does the current technological landscape meet or fall short of Fishman's criteria for sustainable language transmission for children in GME?

My first encounter with a computer was in a GME classroom; however, the digital interface and applications were not built for my language or my culture, and 16 years later, I have a calling to ensure that does not remain the same. Had it not been for my grandparents and the community around me, how could technology have played a role in my active use of idiomatic Gaelic?

2.1.2 Scottish Gaelic - From Educational Exclusion to Digital Inclusion

Gàidhlig (Scottish Gaelic), henceforth referred to as Gaelic, is an autochthonous and both a minority and minoritised language (historic active discouragement of use rather than passive decline). Gaelic was formerly the principal language of the Scottish kingdom until processes of minoritisation began in the twelfth century (McLeod, 2019, pp. 141-142). A clear background of this is brought by (McLeod, 2019, pp. 142-144), who discusses the historical forces that contributed to this minoritisation.

A notable point in history was The (*Education (Scotland) Act*, 1872), where English became the assumed and enforced medium of education, to assimilate Highlanders into an English-speaking world which was seen as a ‘disastrous omission’ by MacLeod (cited in O’Hanlon and Paterson, 2015, p. 304), with The Royal Commission in 1884 condemning the move by saying it was "discouragement and neglect of the native language in the education of Gaelic-speaking children" (cited in O’Hanlon and Paterson, 2015, p. 306)

The exclusion of Gaelic in the past, by institutional means, negatively impacted the language and its community. While the exclusion may not have been driven by an explicit intention to eradicate the language, the absence of its recognition had adverse effects on its spirit and transmission.

This historical precedent suggests that a similar marginalisation could occur in digital contexts. Whether intentional or not, the result of omission or failure to include it appropriately could pose comparable risks, disrupting its linguistic continuity and cultural identity. In a society where everyday linguistic practices increasingly converge around English, one must ask, does Gaelic still occupy an active place in everyday life in Scotland, and if not, what would it mean to lose it completely? Could we reimagine its presence through digital initiatives, and if we don’t act now, how difficult would this task be in the future?

Scotland’s Census found that 2.5% of people over the age of 3 had some skills in Gaelic in 2022. This is an increase of 43,100 people since 2011, when 1.7% had some skills in Gaelic (Scotland’s Census, 2024). Alongside Scots, Gaelic was made an official language of Scotland in the Scottish Languages Act 2025, which received Royal Assent on 1 August 2025 (‘Scottish Languages Act 2025’, 2025). This Act aims to increase support for Gaelic by establishing educational standards (s. 6) and granting Scottish Ministers the duty to promote Gaelic-medium education (s. 8) (‘Scottish Languages Act 2025’, 2025). However, such symbolic recognition may not demonstrate the functional strength of the language, which calls for broader interventions than governmental policies alone.

With the above considered, this project should go to the heart of where language transmission takes place, to the next generation of Gaelic speakers: a class of GME pupils. A deeper understanding of the GME population in this literature review can guide the development of culturally responsive, technological solutions that support learners' engagement with their linguistic and cultural heritage.

2.1.3 GME Student Populations and User Characteristics

Gaelic Medium Education (GME) is a bilingual educational provision where Gaelic serves as the primary language of instruction, aiming to equip students with full fluency and literacy in both Gaelic and English, welcoming students "from families with little or no background in Gaelic" (Education Scotland, 2017, p. 5).

The first Gaelic-medium primary classes appeared in 1985, expanding to a national movement by the 1990s, with the early total immersion model (O'Hanlon and Paterson, 2015, pp. 312-313). The Gaelic Language (Scotland) Act 2005 (*Gaelic Language (Scotland) Act 2005*, 2005) established institutions like Bòrd na Gàidhlig, supporting the Gaelic language and educational planning. By 2013-14, GME was available in 59 primary providers across 14 local authorities (O'Hanlon and Paterson, 2015, p. 313). The diversity of the GME population itself, however, should also be addressed. (McLeod, 2014, p. 8) states: "The language profiles of pupils are equally varied", with many coming from non-Gaelic-speaking homes, especially in urban settings.

O'Hanlon's doctoral research demonstrates that the term Gaelic Medium Education encompasses a vast range of classroom models, with questionnaire data revealing substantial variation in the amount of GME teaching across different subjects at the upper primary level, depending on the school (O'Hanlon, McLeod and Paterson, 2010, p.3). This research identified the P6-7 stage as a critical juncture where Gaelic proficiency begins to lag behind English, with only 20% of P7 students achieving advanced reading levels in Gaelic compared to 48% in English (O'Hanlon, McLeod and Paterson, 2010, p. 27). Moreover, O'Hanlon's survey revealed significant heterogeneity in Gaelic-medium teaching across schools at P7, with some subjects taught predominantly through Gaelic in only 40% of schools, primarily due to difficulties in recruiting specialist teachers fluent in Gaelic (O'Hanlon, McLeod and Paterson, 2010, p. 3).

This inconsistency in immersion, combined with pupils' reduced confidence in Gaelic-medium content and challenges with technical vocabulary at this stage (O'Hanlon, McLeod and Paterson, 2010, pp. 33-35, 39-42), highlights the need for supplementary resources that can provide engaging, natural Gaelic practice outside formal classroom instruction. Digital artifacts represent an effective tool for addressing these gaps, particularly as P6-7 marks the transition to secondary school, where continued Gaelic-medium education becomes increasingly challenging (O'Hanlon, McLeod and Paterson, 2010, pp. 66-67).

O' Hanlon concludes that: "Future research should undertake interviews and other forms of data collection to understand how Gaelic-medium education might draw from and contribute to Gaelic in the community" (O'Hanlon, McLeod and Paterson, 2010, p. 88). This assertion serves as a critical direction for the present project's target user age group, which responds by generating new empirical data focused on the identified age group of P6/7, and by developing a functional prototype to explore how school-based digital interventions can support community Gaelic use.

Figure 1 (page 8) shows the national provision of GME Primary Schools.

Gaelic's evolving linguistic demography, in urban vs rural, home use vs school use, has implications for designing inclusive digital products. The Western Isles of Scotland, where Gaelic is most widely taught, offers a compelling context for investigating how a digital artifact can support language maintenance and cultural continuity.

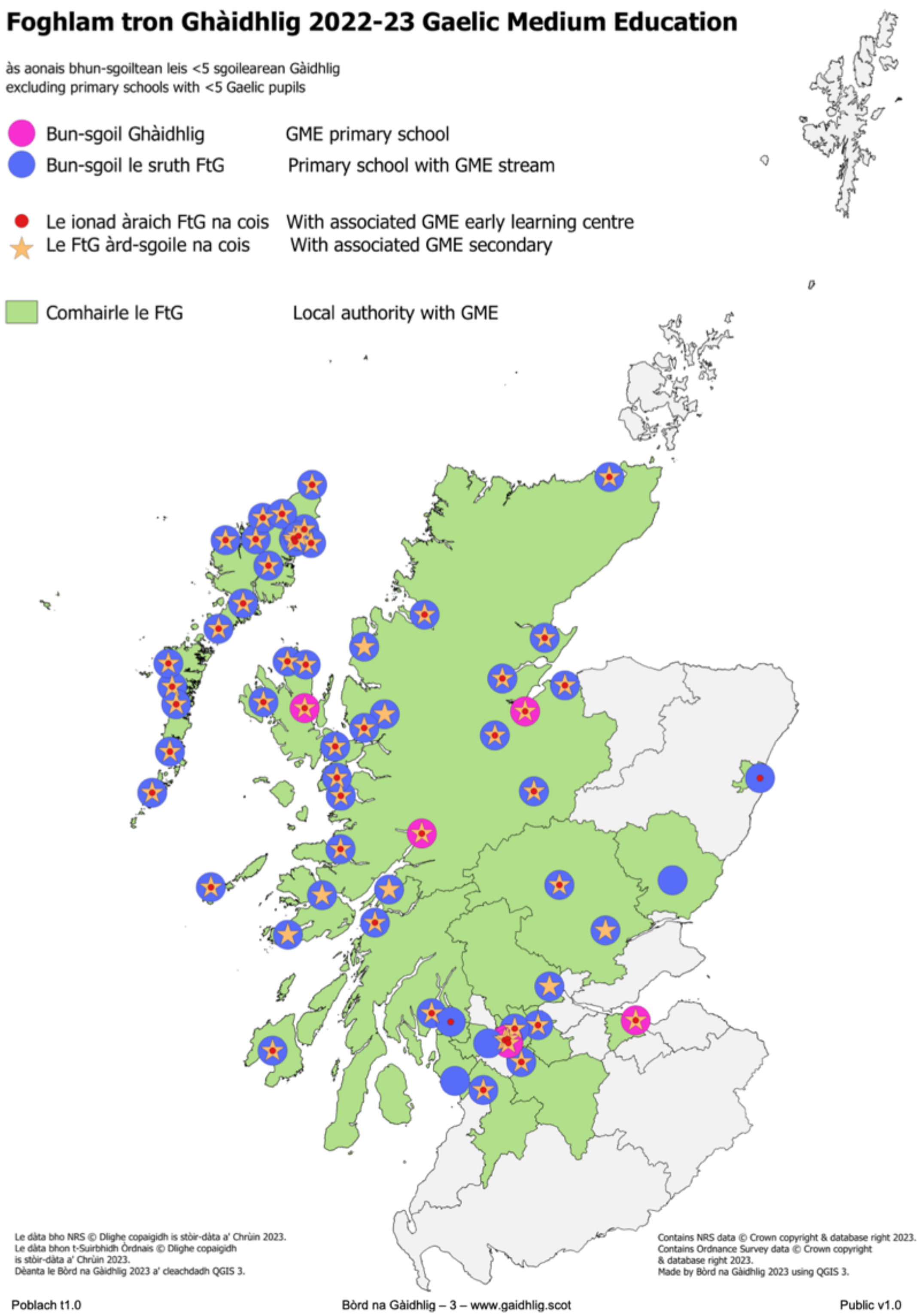


Figure 1: Map of Gaelic Medium Education provision across Scotland in 2022-23, showing primary and secondary schools

2.1.4 Institutional Provision vs. Community Transmission

Although linguistic proficiency can be accomplished in the classroom (Dunmore, 2017, pp. 726-728), the absence of deliberate cultivation of ethnolinguistic identity and positive, supportive language ideologies significantly undermines long-term language retention and community vitality (Dunmore, 2017, pp. 736-739).

As Dunmore argues: "If immersion pupils do not develop social identifications and supportive ideologies toward the languages through which they are educated, it should not necessarily be surprising if they do not then speak the language outside of the classroom, or after completing formal education" (Dunmore, 2017, p. 737).

Again, visiting (Fishman, 1991) framework, successful language revitalisation requires use in the home and community through immersive, informal experiences rather than institutional provision alone (Fishman, 1991, p. 4). These findings highlight a widening gap between institutional provision and community transmission, a tension highly relevant when designing digital environments that seek to preserve authentic Gaelic practice beyond formal schooling.

2.1.5 Idiomatic Gaelic - Empirical Evidence of Cultural Erosion within GME

Sìleas NicLeòid's qualitative study of GME primary school pupils reveals a critical pedagogical gap: idiomatic Gaelic, the language's culturally embedded expressions and nuanced semantic forms, remain systematically absent from classroom linguistic practice (NicLeòid, 2018, pp. 45-61).

Her research, grounded in interviews with pupils, teachers, and parents across GME contexts, demonstrates that most GME-educated children possess limited active and passive knowledge of Gaelic idioms; while pupils develop baseline proficiency in formal classroom Gaelic, they lack exposure to the idiomatic registers and colloquial expressions that characterise authentic community language use. This finding exposes a fundamental misalignment between classroom delivery and communicative authenticity: idioms are rarely integrated as a deliberate educational component in everyday school Gaelic, meaning that pupils rely almost entirely on significant home and community input to develop idiomatic competence, which is a prerequisite for cultural affiliation and sustained language use beyond formal education. The absence of idiomatic Gaelic from GME curricula therefore represents not only a vocabulary gap, but a deeper erosion of cultural and linguistic identity, underscoring how school-based language instruction can inadvertently privilege structural correctness over communicative and cultural authenticity. Gaelic possesses a rich repository of traditional sayings, known as seanfhacail, gnàthasan-cainnt or abairtean, which could be translated as 'old words' or 'proverbs'. Such sayings are often said colloquially in gatherings where the language is used organically in its most authentic form.

The creation of a digital artifact for children in GME invites the opportunity to intentionally integrate this diminishing culture into a game built for digital natives, positioning the game as a vehicle for transmitting culturally rich idiomatic expressions. In doing so, it supports a less stigmatised form of authentic language acquisition, challenging perceptions that such expressions belong solely to older generations, in the hope that a more meaningful and sustainable connection for children engaging with the language is formed. In the scaffolding of a game, embedding these language features within gameplay transforms them from passive museum artifacts into active instruments for meaning making, enabling children to develop communicative competence within authentic sociocultural contexts rather than through decontextualised vocabulary drills.

2.1.5 Previous consideration of Gaelic Games

The potential for Gaelic games has previously been explored and ruled commercially unviable by John Galloway, who conducted over 30 structured interviews with industry practitioners, academics, language technologists, and minority language organisations (Galloway, 2011, p. 3). His exploration was largely affected by financial constraints, given its aim to identify projects offering value for money for Bòrd na Gàidhlig (Galloway, 2011, p. 2). He concluded that minority languages are not considered by commercial game publishers focused on profitable major-language markets (Galloway, 2011, p. 2). He was doubtful about developing original recreational games solely in Gaelic: "It is unlikely that there could be circumstances in which the production of original recreational games in Gaelic alone would be worth considering. The cost would be prohibitive" (Galloway, 2011, p. 8).

Although culturally grounded game development would be the most impactful and appropriate solution for the community, the perceived resource requirements deterred him from this approach (Galloway, 2011, p. 8). Instead, he pivoted to examining the localisation and translation of existing commercial and open-source games (Galloway, 2011, pp. 10-11). After exploring localisation, Galloway stated: "The games companies are focused on profits, which are often uncertain, and will only localise to other languages if the cost and time will be repaid by a significant volume of additional sales. Minority languages are not considered" (Galloway, 2011, Executive Summary). He went on to say, "Not all games lend themselves to localisation: sometimes - as with some Japanese games - they are strongly linked to a cultural context, elements of which are not always easily understood by people in other countries, despite language translations" (Galloway, 2011, , section 2, 'Localisation of games: an overview').

Galloway also states: "Were the localisation of a commercial game to Gaelic feasible, the only requirement of the Gaelic community would be to provide translators for text and, if sound files are necessary, 'actors' for voice-overs". (Galloway, 2011, Executive Summary).

The contribution of the Gaelic community in Gallow's account was limited to translation purposes, which embeds an assumption that technical game development expertise does not reside within the Gaelic-speaking community itself. Despite this report, efforts have been made to create interactive digital artifacts in Gaelic, which will be explored in the next chapter.

Although culturally grounded game development would represent the most impactful and community-appropriate solution, its prohibitive resource requirements have deterred industry implementation. Galloway identified three potentially viable pathways: localisation of commercial games, provision of games on social networking sites, and localisation of open-source games (Galloway, 2011, pp. 10-11).

Yet despite these recommendations, no significant implementation followed. This study addresses a critical methodological gap: although Galloway's analysis focused on economic feasibility, the current research conducts user-centred design evaluation with GME students, generating empirical evidence about design requirements that economic analysis alone cannot address. Notably, Vera Kempe and Ken Scott-Brown of the School of Psychology at the University of Abertay stated they "would be interested to observe the social impact of localised games on the Gaelic community, once they are available" (Galloway, 2011, p9).

2.1.6 Existing Gaelic Games

Several examples of Gaelic games are presented below. While their efforts to integrate the Gaelic language merit recognition, particularly as they establish a digital presence for Gaelic, these initiatives may be characterised as localised or incompletely localised, with translation rather than full integration as their primary approach.

1. **BBC's Game Global Glitch**

Figure 2: Go Jetters: Global Glitch game interface (BBC, accessed October 2025)

Available at:
https://www.bbc.co.uk/games/embed/gplxtqsfmf?exitGameUrl=https%3A%2F%2Fwww.bbc.co.uk%2Fcbeebies%2Fgames%2Fgo-jetters-global-glitch

Available at:https://www.bbc.co.uk/games/embed/gmt5lhq386?exitGameUrl=https%3A%2F%2Fwww.bbc.co.uk%2Fcbeebies%2Fgames%2Falba-go-jetters-crathadh-cruinneil

Figure 3: Go Jetters: Crathadh Cruinneil game interface (BBC, accessed October 2025)

Figures 1 and 2 show an example of a localised game called ‘Global Glitch’. The ‘How to Play’ button in the corner has been translated; however, the name of the game itself has remained the same.

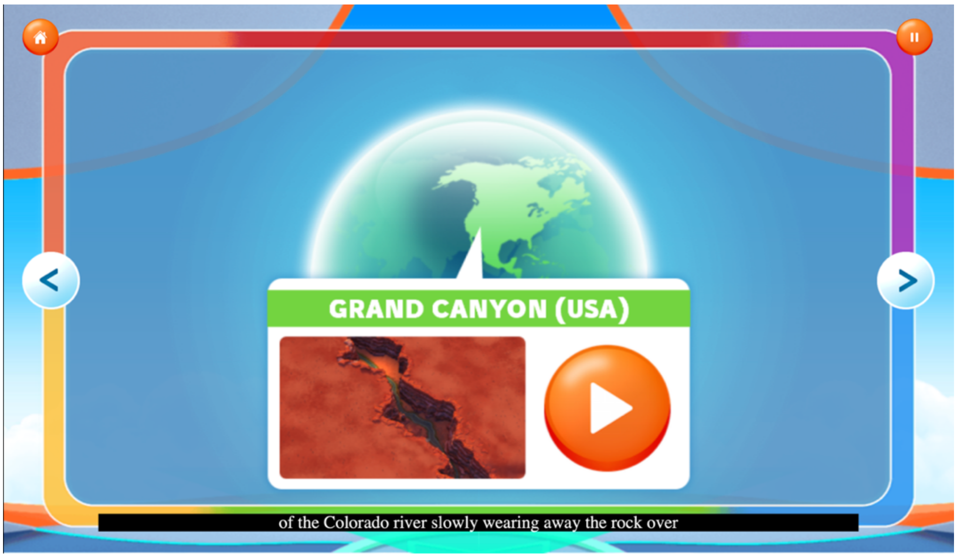


Figure 4: Go Jetters: Global Glitch gameplay (BBC, accessed October 2025)

Figure 5: Go Jetters: Crathadh Cruinneil gameplay (BBC, accessed October 2025)

Gameplay with audio recordings, labels, and subtitles was all provided in Gaelic, in a native style of language. This is an example of the localisation an English BBC Game.

2. Go Gaelic’s Language Reinforcement Mini-Games

Figure 6: GoGaelic Gameplay 1

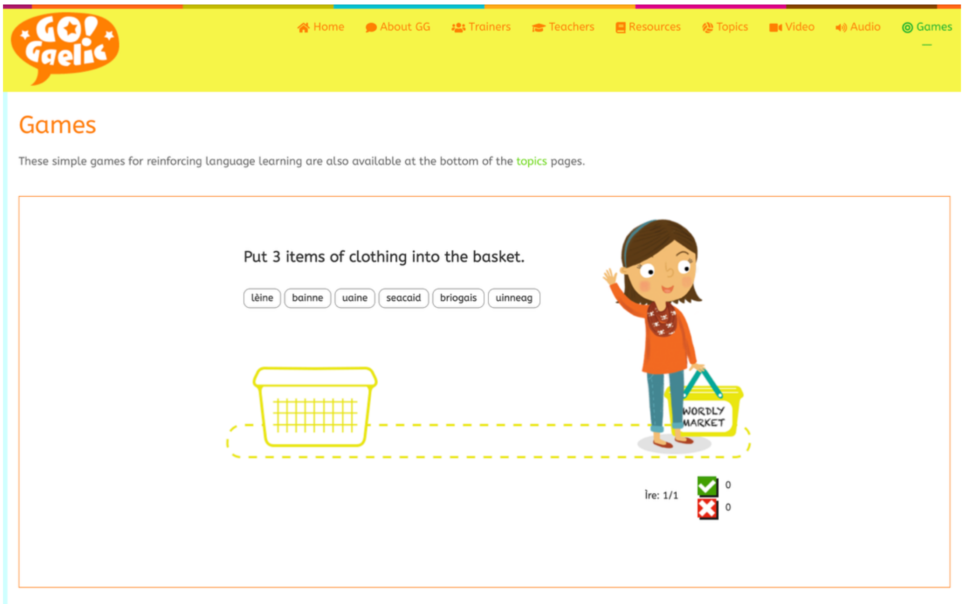


Figure 7: GoGaelic Gameplay 2

Figure 6 shows English Instruction with Gaelic keywords, and Figure 7 also incorporates English in its answers. This game was created solely for Gaelic learners and it's the type of platform which a user may only use once, in the very early stages of their Gaelic leaning journey, and may be seen as unsuitable for the GME students themselves if they do not yet know how to read English.

2.2 Learning from Parallel Contexts - Cross-Culture Design

Although Gaelic has limited work in digital games that are culturally responsive, substantial insights can be drawn from cross-cultural design approaches and parallel contexts where games serve as vehicles for cultural and linguistic preservation. By examining the design principles, user research methodologies, and implementation strategies from other minority language communities, we can establish evidence-based frameworks applicable to Gaelic language education.

2.2.1 Cultural Elements - Appropriate & Authentic Design

A core design challenge is the appropriate representation of language and culture, especially when designing elements for minority language games. Zhang differentiates indexical and iconic elements (Zhang *et al.*, 2025). Indexical cultural elements are described as historically accurate, directly referential cultural content. Whereas iconic cultural elements are symbolically representative and creatively adapted representations of culture.

Typical indicators of game success are hedonic values such as enjoyment or motivation, but limited attention is paid to games as cultural products. Zhang investigated the impact of Chinese cultural elements on player evaluations of games (Zhang *et al.*, 2025).

Zhang identified three critical findings. First, while indexical elements significantly enhance perceived cultural authenticity, this authenticity does not automatically improve overall game evaluations, a counterintuitive result suggesting that historical accuracy alone is insufficient for positive player reception. Second, iconic elements demonstrate the opposite pattern: they increase perceived design effort, which directly enhances game evaluations, indicating that creative cultural adaptation is more effective for player satisfaction than strict historical fidelity. Third, the effect of indexical elements is conditional on player motivation: authenticity-seeking players rate indexical elements favourably, whereas other player segments may find them mismatched to their expectations (Zhang *et al.*, 2025).

An informative finding was the backfire effect of authenticity. High cultural authenticity was found to detract from game evaluations for most players. Indexical elements could be seen to disrupt the immersive, virtual nature of games by grounding them too heavily in reality/history. Players who seek escapism may find overly realistic cultural elements reduce their sense of immersion.

This creates some design trade-offs. Using authentic Gaelic cultural elements (indexical) may enhance cultural authenticity but not necessarily improve user evaluations or engagement. Gaelic-inspired elements (iconic) that are adapted for digital contexts may be more effective for broader audiences. There is no strict rule to have one and not the other; however, as this game is invested in Gaelic's cultural authenticity, indexical elements shall be considered.

Designing with a purely indexical approach, although culturally authentic, risks making gameplay historically restraining, making it feel like a museum experience rather than a lived cultural space, and this could alienate GME pupils who have a contemporary, not historical, relationship with the language. A hybrid approach allows both casual engagement and deeper cultural exploration. The design stage will choose a balance between indexical and iconic elements appropriately, although following the gap in idiomatic Gaelic for children of the P6/7 age group, seanfhacail / proverbs will be considered as the most appropriate indexical part of the artifact.

2.2.2 Balancing Content and Play

Designing digital environments for true cultural interaction requires strategically balancing digital games as content with digital play experiences that facilitate cultural immersion. A distinction between digital games (DG) and digital play (DP) is introduced by (Veresov and Veraksa, 2023, p. 1092) where they state: digital games should be seen as "material/virtual content having a purpose and objective, pre-determined settings, rules, steps, actors, etc." whilst digital play as "a system of rules, roles, narratives and play actions".

This distinction is important for this game's development as it highlights that the game content (DG) must contain appropriate mechanics and storylines, which then facilitate meaningful play activities (DP) that immerse children in the Gaelic language and cultural contexts.

This supports the argument that a game built in English and subsequently translated into Gaelic cannot function as a culturally responsive tool for language transmission, as it does not consist of appropriate DP elements to transmit the cultural element of the language, the mechanism through which language acquisition occurs naturally and meaningfully.

2.2.3 What does normalisation look like?

(Veresov and Veraksa, 2023, p. 1093) go on to say: "Culture might be understood as a system of 'normative situations'. Normative situations are repetitive; they stabilise the social interaction of members of the society precisely by means of the prescriptions". In a Gaelic context, this could mean creating culturally authentic scenarios that reflect broader cultural lifestyles and traditions. A key insight from (Veresov and Veraksa, 2023, p. 1093) : "An object and a rule create a cultural situation with a prescribed norm of behaviour - a normative situation…Normative situations in a child's everyday life, therefore, involve both natural and cultural aspects that shape the actions of a child."

A Gaelic game could incorporate normative situations authentic to Gaelic culture, be it linguistically using sayings to match a situation where they'd be accurately used contextually or through familiar landscape settings such as a croft, beach, or town centre, if designing for rural dwellers, or even traditional activities like ceilidhs, fishing, or community gatherings where specific social rules and language patterns apply. Through this, players can internalise both language patterns and culture-specific social rules.

Veresov and Veraksa state: "The more variety of normative situations (and corresponding play actions) the game provides, from less complex to more complex ones, the better it contributes to the child's development. However, these situations must reflect the real cultural norms of life, i.e. human cultural norms." (Veresov and Veraksa, 2023, p. 1095). In the context of this game, normalisation of traditional or historic activities may be alienating to some members of the Gaelic community, as perhaps their families or friends do not participate in traditional activities like fishing or ceilidhs, rather the game should feel intrinsically accessible to all players without a need to go 'back in time' to understand how the game works. My aim to design with cultural authenticity can be achieved through designing situations that appropriately use idiomatic Gaelic and contain familiar elements that children may resonate with locally, instead of generic scenarios with Gaelic labels.

2.2.4 "Nothing about us, without us."

A 2022 paper written by Harbord, Lyons, and Dempster provides critical insights into ethical considerations when designing digital games to preserve indigenous languages & traditions. The phrase "nothing about us, without us" serves as a powerful ethical principle, emphasising that indigenous people should be involved throughout any game's development, not as translators or fact checkers. In the paper, an Indigenous Cultural Advisor from the Mississaugas of Scugog Island First Nation stated: "I have heard many times, that if a language dies off, so does the culture. Our languages are tied so closely with our culture, I believe this to be true" (Harbord, Lyons and Dempster, 2022, p.5). Those comments further emphasise the need for localised games, embedded with culturally appropriate, normative elements and situations with idiomatic speech rather than direct translations of games built for other generic user groups.

A critical takeaway from a Sioux Lakota game designer: not all cultural knowledge should be gamified. He said: "There's so many in our culture, they're tied directly to our spirituality, that I would never want to put in a game" (Harbord, Lyons and Dempster, 2022, p.6). This insight is a reminder to ensure idioms or other game elements are presented with sufficient cultural context to honour their significance, not as decontextualised vocabulary, to make the culture 'fit' into an ideal.

Another finding was that most indigenous-designed games are created as non-profit products, often available for free, with only the aim of sharing or preserving heritage, not generating revenue. The paper notes: "It is this level of altruism that has sometimes stopped these kinds of games from becoming huge mega-hits. As the budget is not there to create the mass media promotion and hype needed to become the bestselling games" (Harbord, Lyons and Dempster, 2022, p.14). While Galloway (2011) identified this financial constraint, his report framed it as a barrier to viability, rather than exploring alternative, non-commercial models from academia or talent within the community itself.

2.2.5 User Centred Design in Practice for a Minority Language

I look at an effort that took place at Institut Teknologi Bandung, where they created a local-language learning app to address minority-language endangerment. The study employs ISO 9241-210's four-stage user-centred design process, which demonstrates how formal UCD methodology can be adapted to minority language contexts (Felicya and Lestari, 2024, p. 1). The final prototype achieved high usability and user experience scores - an average SEQ score of 6.93 out of 7 and an average SUS of 95 out of 100 (Felicya and Lestari, 2024, p. 1), demonstrating the effectiveness of this approach in supporting language revitalisation efforts.

Their incorporation of avatars based on Indonesian exotic animals reinforced regional cultural identity and engagement: "The avatars, which represent students in the app, are chosen from Indonesia's exotic animals" (Felicya and Lestari, 2024, p. 5). By choosing animals that are resonant with those in the GME community, the users can feel a closer representation of their surrounding world, further enhancing the normative situations mentioned previously in this section.

Strong colour psychology was used to reinforce the desired emotional and motivational state in users: "The dominant color used is yellow, complemented by brown and gold, which reflect the traditional culture of the region. In color psychology, yellow is considered a color that can stimulate brain and mental activity and gives an aura of warmth, optimism, and enthusiasm" (Felicya and Lestari, 2024, p. 4). This also provides a prompt to explore colours, patterns, or materials that closely resonate with those in the GME community.

Felicya & Lestari’s choice to use the user-centred design methodology is a parallel with my own efforts to ensure the design of my artifact is well-suited to its users.

2.3 Game Design

This section outlines the theoretical frameworks and empirical evidence that inform design decisions. The MDA (Mechanics-Dynamics-Aesthetics) framework provides a structured approach to aligning interactive systems with intended outcomes. The section then examines how balancing content, challenge, and narrative sustains engagement across multiple interactions. Finally, it considers research demonstrating that active participation over passive consumption drives engagement in child-oriented interactive experiences, informing the prioritisation of dynamic and exploratory elements in the design.

2.3.1 MDA - Theoretical Framework:

The MDA framework consists of three interconnected layers that represent different levels of abstraction in game design: Mechanics, Dynamics, and Aesthetics. (Hunicke, LeBlanc and Zubek, 2004, p. 2)

- **Mechanics** are the underlying systems' rules, algorithms, data structures, and controls that players interact with directly.
- **Dynamics** describe the emergent behaviour during gameplay, arising from mechanics interacting with player input over time.
- **Aesthetics** describes the emotional responses and experiences the game evokes. MDA defines eight categories: Sensation, Fantasy, Narrative, Challenge, Fellowship, Discovery, Expression, and Submission.

A fundamental insight from the MDA framework is that a game's core content is its dynamic behaviour, not media streaming toward players. This framing treats games as interactive systems rather than passive media delivery mechanisms.

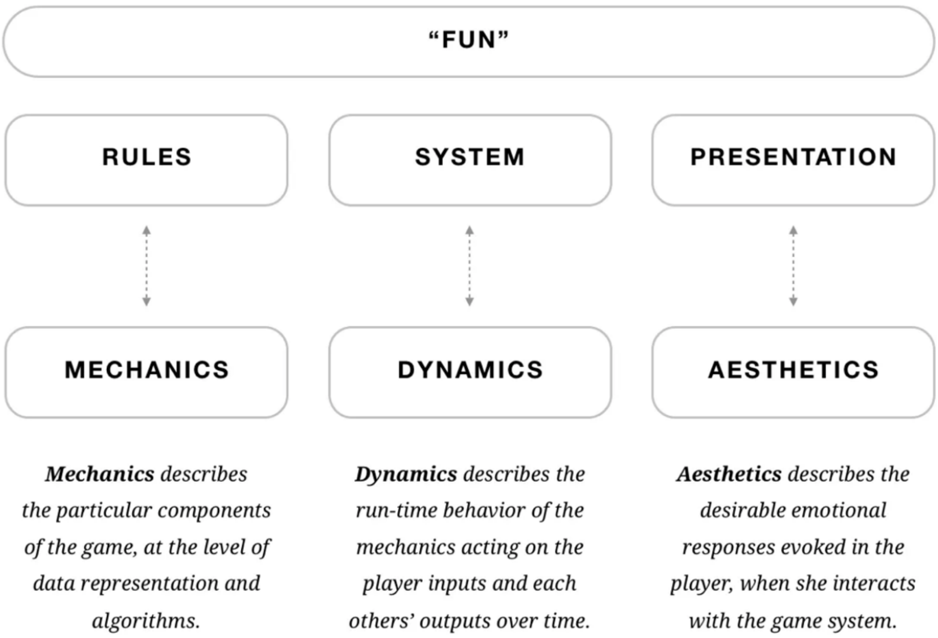


Figure 8: 'Using the MDA Framework as an Approach to Game Design', (Carroll, J. (2013)

Available at: https://medium.com/@jenny_carroll/using-the-mda-framework-as-an-approach-to-game-design-9568569cb7d (Accessed: October 2025)

2.3.2 Balancing Content, Challenge, and Narrative

Wang distinguishes between games designed for one-time play, characterised by narrative-driven experiences resembling novels or films where discovery and immersion form the main function, and games designed for repeated play, where engagement could come from competing against multiple opponents who each contribute different strategies or skills (Wang, 2023, pp. 34-35). For educational games intended to incentivise multiple playthroughs, designers could employ procedural variation through randomised encounters or evolving challenge structures or implement adaptive difficulty mechanisms that dynamically respond to player performance, thereby sustaining engagement beyond the initial playthrough (Wang, 2023, p. 35).

Wang's guide to using multiple playthroughs and evolving challenge structures will be considered to ensure the game is not purely static or for one use. Should I want to incentivise multiple plays, I should explore procedural variation (different encounters, evolving challenges) and adaptive difficulty to sustain engagement beyond initial play. This could be achieved by creating some mini-games within the game that require different problem-solving abilities.

2.3.3 Active Participation Over Passive Consumption as a Driver of Child Engagement

(Rodríguez *et al.*, 2020) conducted a between-subjects experimental study examining the effect of gamification on digital Cultural Probes (CPs), a user research method enabling participants to self-report information through tasks completed at their own pace, with children aged 11-12 years. 25 participants from two primary schools in Barcelona interacted with either a gamified or non-gamified version of a digital Cultural Probe Android application across four distinct tasks. The gamified version incorporated narrative and role-play elements, missions, rewards, competition mechanics, and time constraints (Rodríguez *et al.*, 2020).

The study revealed that interactive and dynamic tasks generated the highest levels of child engagement. The "Electrician task," which required children to identify electrical appliances across different rooms, emerged as the most favoured activity because "it was the one that required the most input from the children" (Rodríguez et al., 2020, Section 5.5). Conversely, the "Journalist task," which involved conducting structured interviews, was the least popular due to its extensive reading requirements and perceived length (Rodríguez *et al.*, 2020, Section 5.5).

While the gamified version demonstrated marginally higher engagement levels compared to the non-gamified control condition, these differences did not achieve statistical significance. (Rodríguez *et al.*, 2020, Section 5.5) attribute this outcome to the limited number of tasks and the study's short duration, positing that gamification may yield more substantial benefits in contexts involving longer-term engagement periods or a greater number of tasks. Critically, the study concluded that "most children prefer dynamic and interactive tasks to those that are more cognitively intensive, involving reading and understanding" (Rodríguez *et al.*, 2020, Section 5.5). This finding supports the design decision to avoid large text during games and instead prioritise dynamic and interactive tasks.

2.4 Literature Review Summary

The following summary connects the three major sections of this literature review, with each section summary distilling the key theoretical foundations and empirical evidence that establish the critical need for this project, the design principles guiding its implementation, and the game design frameworks that will operationalise these principles into a functional artifact.

Section 2.1: Gaelic - A Language Between Tradition and Technology

This section established the foundational rationale for digital intervention in GME contexts by exposing a critical transmission gap that technology could address. Fishman's framework (Section 2.1) demonstrates that sustainable language revitalisation depends fundamentally on "intergenerational and demographically concentrated home-family-neighbourhood" transmission, yet contemporary GME contexts increasingly lack this foundation, with many pupils coming from non-Gaelic-speaking households where the informal transmission mechanisms Fishman identifies remain inaccessible.

NicLeòid's empirical research (Section 2.1.5) reveals the pedagogical consequence: idiomatic Gaelic - the seanfhacail, gnàthasan-cainnt, and colloquial registers characterising authentic community language use - remains systematically absent from GME classrooms, with pupils developing baseline structural proficiency while lacking exposure to the culturally embedded expressions that enable genuine communicative authenticity. This absence represents not only vocabulary deficiency but cultural and linguistic identity erosion, as classroom instruction privileges grammatical correctness over the communicative and cultural authenticity prerequisite for sustained language use beyond formal education. O'Hanlon's longitudinal study (Section 2.1.3) identifies P6-P7 as the critical juncture where Gaelic proficiency begins lagging behind English, with only 20% of P7 students achieving advanced Gaelic reading levels compared to 48% in English, compounded by significant heterogeneity in Gaelic-medium teaching provision and pupils' reduced confidence in Gaelic-medium content at this transitional stage before secondary school.

Dunmore's research (Section 2.1.4) reinforces that linguistic proficiency alone proves insufficient; without deliberate cultivation of ethnolinguistic identity and supportive language ideologies, immersion pupils often fail to speak the language outside classroom contexts or after completing formal education.

Galloway's commercial viability assessment of Gaelic games (Section 2.1.5, 2.1.6) concluded that minority languages are dismissed by profit-focused publishers, with game localisation proving culturally inappropriate when titles remain "strongly linked to" source cultural contexts, yet his framework limited community contribution to translation roles and dismissed purpose-built development as prohibitively expensive, failing to explore academic-community partnership models.

Existing Gaelic digital games (Section 2.1.6) demonstrate translation-focused approaches - BBC's partially localised titles retain English game names despite Gaelic gameplay elements, while Go Gaelic's learning platforms incorporate English instructions unsuitable for GME pupils with limited English literacy, establishing that current provision fails to meet the community's needs for culturally grounded, purpose-built artifacts.

These converging findings establish the project's imperative: develop a digital game intentionally integrating age-appropriate idiomatic expressions within culturally normative contexts, targeting the P6-P7 demographic with a purpose-built artifact designed to cultivate both linguistic competence and ethnolinguistic identity as a scalable proxy for the authentic home-family-neighbourhood transmission Fishman identifies as essential, thereby addressing the documented gap between institutional provision and community transmission while responding to the specific needs of upper primary pupils at this critical educational juncture.

Section 2.2: Learning from Parallel Contexts - Cross-Cultural Design

This section synthesises design principles and ethical frameworks from parallel minority language contexts to establish evidence-based methodologies for culturally authentic digital artifact development. Zhang et al.'s empirical research (Section 2.2.1) distinguishes indexical elements (historically accurate, directly referential cultural content) from iconic elements (symbolically representative, creatively adapted cultural representations), revealing a counterintuitive "backfire effect": while indexical elements enhance perceived cultural authenticity, high historical accuracy can detract from game evaluations by grounding gameplay too heavily in reality, creating museum-like experiences alienating players seeking contemporary engagement, whereas iconic elements increase perceived design effort and directly enhance player satisfaction, suggesting that creative cultural adaptation proves more effective than strict historical fidelity. This necessitates a hybrid design approach where indexical elements transmit authentic linguistic and cultural knowledge - specifically, by embedding seanfhacail that address NicLeòid's identified gap, alongside culturally authentic settings, linking language to lived experience - while iconic elements maintain contemporary accessibility through adapted game mechanics and visual representations, ensuring cultural authenticity without alienating GME pupils who have contemporary rather than historical relationships with Gaelic.

Veresov and Veraksa's theoretical framework (Section 2.2.2, 2.2.3) provides crucial scaffolding by distinguishing digital games (predetermined content with rules, objectives, and settings) from digital play (systems of rules, roles, and narratives facilitating meaningful activity), demonstrating that translation-only approaches fail because games built in English subsequently translated lack culturally appropriate mechanics through which language acquisition occurs naturally and meaningfully. Their concept of "normative situations" - repetitive scenarios stabilising social interaction through prescribed cultural norms where "an object and a rule create a cultural situation" enabling children to internalise both language patterns and culture-specific social rules - requires designing varied contexts (from simple to complex) reflecting real cultural practices rather than generic scenarios with Gaelic labels, while carefully avoiding overly traditional activities that may alienate pupils whose families don't participate in activities, instead focusing on universally accessible local elements and familiar landscape features that resonate with island-dwelling pupils' lived experiences.

Harbord, Lyons, and Dempster's indigenous design research (Section 2.2.4) establishes the ethical principle "nothing about us, without us," emphasising that indigenous people must be involved throughout development rather than relegated to translation roles, while cautioning that not all cultural knowledge should be gamified (particularly spiritual/sacred elements) and recognising that most indigenous-designed games operate as non-profit heritage transmission projects, prioritising cultural preservation over commercial viability.

Felicya and Lestari's successful minority language app development (Section 2.2.5) demonstrates ISO 9241-210 UCD methodology achieving 95/100 SUS scores through strategic design choices: incorporating local animal avatars (Indonesian exotic animals) to reinforce regional cultural identity while circumventing human representation biases, which is particularly important given users' varied ethnicities. The convergence of these frameworks directs the artifact toward employing non-human characters grounded in locally familiar Hebridean animals to avoid representational essentialisms, establishing a 100% Gaelic linguistic environment with no English fallbacks to position Gaelic as the normative interaction language. By embedding idiomatic expressions contextually within normative situations reflecting authentic island familiar community practices or surroundings, creates an artifact that transmits both language and cultural identity through an ethically grounded and community-centred design process.

Section 2.3: Game Design

This section establishes the theoretical frameworks and empirical evidence base for translating cultural and linguistic design principles into functional gameplay systems that sustain engagement while facilitating authentic language transmission. The MDA (Mechanics-Dynamics-Aesthetics) framework (Section 2.3.1) provides structured methodology for aligning gameplay systems with desired player experiences across three interconnected layers: Mechanics (underlying rules, algorithms, data structures, and controls players interact with directly), Dynamics (emergent behaviour during gameplay arising from mechanics interacting with player input over time), and Aesthetics (emotional responses and experiences the game evokes across eight categories including Sensation, Fantasy, Narrative, Challenge, Fellowship, Discovery, Expression, and Submission), with the fundamental insight that a game's core content emerges from dynamic behaviour rather than passive media streaming, treating games as interactive systems whose mechanics must be designed to generate the specific dynamics and aesthetics that support language learning objectives while maintaining player engagement.

Wang's replayability research (Section 2.3.2) distinguishes games designed for one-time narrative-driven play from those designed for repeated engagement, establishing that educational games intended to incentivise multiple playthroughs require procedural variation through randomised encounters or evolving challenge structures, or adaptive difficulty mechanisms dynamically responding to player performance to sustain engagement beyond initial play, thereby ensuring the artifact functions as an ongoing language practice tool rather than a novelty experiencing rapid engagement decay after first use.

Rodríguez et al.'s experimental study with children aged 11-12 (Section 2.3.3) reveals that interactive and dynamic tasks requiring substantial player input generate significantly higher engagement than intensive text-heavy tasks, with their finding that "most children prefer dynamic and interactive tasks to those that are more cognitively intensive, involving reading and understanding" directly supporting the design decision to prioritise mechanics requiring active player input (clicking, spatial reasoning, memory challenge, strategic placement) over reading-intensive activities, utilising character-driven dialogue and visual cues to convey instructions and narrative progression rather than text blocks that impose cognitive load while potentially disrupting the immersive linguistic environment.

Literature Review Conclusion

These frameworks collectively direct the game toward implementing mechanics that generate challenge-based and discovery-oriented aesthetics through spatial puzzles or memory tasks. The design incorporates difficulty progression and randomised elements to incentivise repeated engagement. Text-heavy instructional content is minimised in favour of tutorial systems embedded within gameplay, delivered through guided first experiences or character-mediated instructions. All game elements, from interface interactions to feedback systems and progression mechanics, must operate exclusively in Gaelic, establishing the language as the natural medium of interaction. In doing so, the game operationalises the theoretical insights from Sections 2.1 and 2.2 into concrete design specifications that balance educational objectives (idiomatic language transmission and ethnolinguistic identity cultivation) with engagement imperatives (dynamic interaction, sustained replayability, and age-appropriate challenge progression). Finally, cultural authenticity is maintained through normative situations and hybrid indexical/iconic representation strategies, positioning the artefact as both pedagogically effective and intrinsically motivating for the P6-P7 target demographic. This design focus aligns with the critical juncture at which supplementary digital resources can meaningfully address documented gaps in Gaelic Medium Education provision and authentic community transmission identified throughout this literature review.

Chapter 3. Methodology

Bidh an t-eun a' seinn mar a thogadh e *Tobar an Dualchais, Leòdhas*	The bird sings as it was raised *Tobar an Dualchais, Lewis*

This chapter connects the topics discussed in chapter 2 with the aims of this research, whilst adhering to practical constraints intrinsic to this project's timeline and educational research involving children. Drawing on ISO 9241-210's human-centred design principles, the research adopts a pragmatic mixed-methods approach that balances iterative design ideals with the realities of limited school access. The comprehensive literature synthesis in Chapter 2 is treated as the primary source of insight into user requirements, particularly regarding Gaelic, cultural authenticity, and digital normalisation. My own background as a former GME pupil, as well as stakeholder communication throughout development, will further supplement the game's Gaelic-first design and its requirements.

3.1 Research Aims & Questions

This research aims to design, develop, and evaluate a culturally grounded Gaelic digital game that supports active language use among upper primary Gaelic Medium Education (GME) students. The research questions guiding this study reflect two modes of inference:

- **Induction** (Learning from empirical observation (Thomas, 2003, p. 3)
- **Abduction** (Generating novel explanations for surprising observations (Timmermans and Tavory, 2012, p. 169-171)

3.1.1 Research Questions

The methodological approach is designed to answer the following core research questions:

(Q1) Usability: To what extent is the game prototype usable for GME students aged 8-12 with minimal prior instruction?

(Q2) Engagement & Experience: How do students and staff experience the game, and what are their perceptions of its cultural authenticity and engagement value?

(Q3) Language Use: In what ways does incorporating idiomatic Gaelic in the game support authentic language usage during play?

(Q4) Emergent Phenomena: What unexpected barriers, facilitators, or phenomena emerge that were not anticipated by the design, and what plausible explanations might account for them?

Table 2: Research Question 1

Induction	**(Q1) Usability:** To what extent is the game prototype usable for GME students aged 8-12 with minimal prior instruction?
	This question focuses on basic operability and efficiency of interaction, aligned with ISO 9241-11's definitions of effectiveness and efficiency, in a real GME classroom context. Addressed by: • Behavioural metrics: task completion time and error rates. • Questionnaire data: self-reported satisfaction using an adapted System Usability Scale (SUS).

Table 3: Research Question 2

Induction	**(Q2) Engagement & Experience:** How do students and staff experience the game, and what are their perceptions of its cultural authenticity and engagement value?
	This question reflects the dual emphasis, identified in Chapter 2, on hedonic engagement and the cultural suitability and authenticity of indexical and iconic elements in minority-language game design. Addressed by: • Pupil discussion group: cultural interests, game habits, and perceptions of cultural elements. • Pupil questionnaire: self-reported engagement and satisfaction. • Teacher questionnaire: professional evaluations of engagement and cultural fit.

Table 4: Research Question 3

<table>
<tr><td rowspan="2">Induction</td><td>(Q3) Language Use: To what extent do pupils and teachers perceive Gaelic as a language of gameplay (i.e. understandable, useful, and appropriate in this context), and under what conditions would pupils choose to use Gaelic in games by preference rather than default?”</td></tr>
<tr><td>This question directly responds to the documented gap in idiomatic Gaelic in GME classrooms discussed in Section 2, examining whether the artefact can act as a vehicle for idiomatic, culturally grounded language use.

Addressed by:
Pupil questionnaire items on language use and perceived authenticity of idiomatic expressions.
Discussion group prompts on game and language preferences.
Observational field notes on spontaneous Gaelic use, and language choice in peer interaction.</td></tr>
</table>

Table 5: Research Question 4

<table>
<tr><td rowspan="2">Abduction</td><td>(Q4) Emergent Phenomena: What unexpected barriers, facilitators, or phenomena emerge that were not anticipated by the design, and what plausible explanations might account for them?</td></tr>
<tr><td>Addressed by:
Pupil discussion group reflections.
Observational field notes capturing unanticipated patterns in interaction, engagement, or language use.
Open-ended teacher questionnaire responses.</td></tr>
</table>

3.1.2 Research Scope & Boundaries

The study is not designed as a formal assessment of Gaelic proficiency or of pupils' general technical or problem-solving abilities. Gaelic fluency and digital skills are treated as background characteristics inherent in the GME population rather than as experimental variables. The focus is on how well the game:

- Affords usable interaction for the typical age group of GME pupils
- Supports engaging, culturally resonant play
- Provides opportunities for authentic, idiomatic Gaelic use during gameplay

Accordingly, findings are interpreted as insights into the interaction between the artefact and the user group in context, rather than as generalisable measures of individual ability.

3.1.3 Research Paradigm and Design

This study adopts a pragmatist research paradigm, which emphasises selecting methods based on their practical usefulness in addressing the research questions rather than adherence to a single epistemological tradition (Johnson and Onwuegbuzie, 2004, pp. 17-18). Pragmatism is well-suited to applied HCI and educational research where the primary objective is to understand "what works" in a particular context and why (Johnson and Onwuegbuzie, 2004, pp. 17-18).

As outlined in Chapter 2, game-based interventions in minority language contexts implicate usability, user experience, cultural authenticity, and linguistic behaviour simultaneously. No single method can adequately capture this complexity. To address this, the evaluation phase uses a convergent parallel mixed-methods design (John W. Creswell and Vicki L Plano Clark, 2011, pp. 69-70):

- Quantitative and qualitative data are collected concurrently during a single school-based research session (John W. Creswell and Vicki L Plano Clark, 2011, p. 64)
- Each strand is analysed separately using appropriate techniques (John W. Creswell and Vicki L Plano Clark, 2011, p. 66)
- Findings are then integrated during interpretation, allowing qualitative insights (the "why") to contextualise quantitative patterns (the "what") (John W. Creswell and Vicki L Plano Clark, 2011, p. 70)

3.2 Design Frameworks

The artefact's development is structured around two complementary design stages:

- A User-Centred Design (UCD) stage, where user needs, contexts, and cultural environments are understood and translated into game design choices.
- A Game Design Document (GDD) stage, where these requirements are concretised into specific mechanics, dynamics, aesthetics, and content.

Together, these stages ensure that the game is both technically robust and culturally grounded, responding to the need for culturally responsive digital artefacts in GME identified in Chapter 2.

3.2.1 User Centred Design (UCD)

ISO 9241-210 specifies four core human-centred design activities:

- Understanding and specifying the context of use.
- Specifying user requirements.
- Generating and refining design solutions.
- Evaluating designs against requirements.

Traditional UCD presumes multiple iterative cycles across these activities, with sustained user involvement. In the present study, full iteration testing is constrained by single-session school access, but the spirit of UCD is maintained by front-loading contextual understanding through a comprehensive literature review and stakeholder involvement during design.

The context of use for this artefact includes the technical, physical, social, cultural, and organisational environments in which GME pupils interact with digital tools, consistent with ISO 9241-11's definition of environment. Given the specific challenges of Gaelic digital normalisation and idiomatic language transmission outlined in Sections 2.1-2.2, particular attention is paid to:

- Pupils' varied home and school language practices
- Sociocultural expectations around Gaelic use
- The classroom and device infrastructure typical of GME settings

UCD Data Gathering

UCD data gathering for this project draws on multiple sources:

- The **Literature Review** (Chapter 2), which synthesises findings on GME demographics, idiomatic Gaelic, cultural authenticity in game design, and minority-language digital tools.
- A **User Group Profile**, exploring the characteristics of P6-P7 GME pupils and their learning context.
- **Personas** that embody representative user types and their goals, informed by literature and stakeholder insights.
- **Reflexive Design** which discusses how my background as a former GME pupil and local of the Outer Hebrides informed the design process. Drawing on my lived familiarity with Gaelic language practices and cultural norms, I use reflexivity to ensure the game's design authentically represents idiomatic and community-based Gaelic use.
- **Stakeholder communication**, particularly with GME teachers, to ensure alignment with classroom realities and cultural expectations, reflecting the "nothing about us, without us" principle discussed in Section 2.2.4.

These inputs are synthesised into a set of functional and non-functional requirements that guide the subsequent GDD.

3.3.2 GDD: The Game Design Document

The Game Design Document (GDD) serves as the primary design specification and communication tool throughout the project lifecycle. It translates UCD-derived requirements into concrete game elements, including:

- Core mechanics and rules
- Narrative framing and challenge structure, informed by the MDA framework discussed in Section 2.3
- Integration of idiomatic Gaelic expressions at appropriate points in the gameplay loop
- Visual and interactive elements that balance indexical and iconic cultural representations, as theorised in Section 2.2.1.

The GDD thus functions as a bridge between high-level design motivations (e.g. digital normalisation, idiomatic language use) and the implemented artefact, ensuring coherence between theoretical insights and technical implementation.

3.4 Development Framework

Water-Scrum-Fall is a hybrid development methodology that effectively allows planning and agility according to the structural constraints inherent to this academic project. This approach combines the upfront documentation and linear sequencing of 'Waterfall' with the iterative refinement capabilities of 'Scrum', an approach that has become the empirical norm in modern software development rather than a theoretical compromise (Schwaber, 1997).

For a Senior Honours project with fixed academic deadlines, URTEC approval requirements, a mid-way flexible development period, and a single access window to the research site for data collection and evaluation, a three-stage process is naturally configured, and Water-Scrum-Fall offers a defensible methodological foundation that reconciles the demands of rigorous upfront planning, iterative development, and delivery

"In 2011, Dave West coined the term Water-Scrum-Fall…West uses the term *water* to describe the upfront work done by the organisation in doing feasibility studies, funding exercises, and requirements gathering. West uses the term *scrum* to describe the middle proces where developers use the Scrum methodology to do actual development (and possible testing). And finally, West uses the term *fall* to describe the final deployment step where the organisation's existing release policy is still implemented." (Butgereit, 2018, p. 177). Water-Scrum-Fall provides a pragmatic middle ground for projects that, due to external or internal constraints, cannot support multiple full feedback cycles and unrestricted iterative releases.

Rather than attempting a pure Agile approach that would require continuous user interaction, proving infeasible in this educational context, Water-Scrum-Fall allows the project to establish structural clarity during planning phases while preserving meaningful iterative refinement during development.

Water Scrum Fall applied to this project:

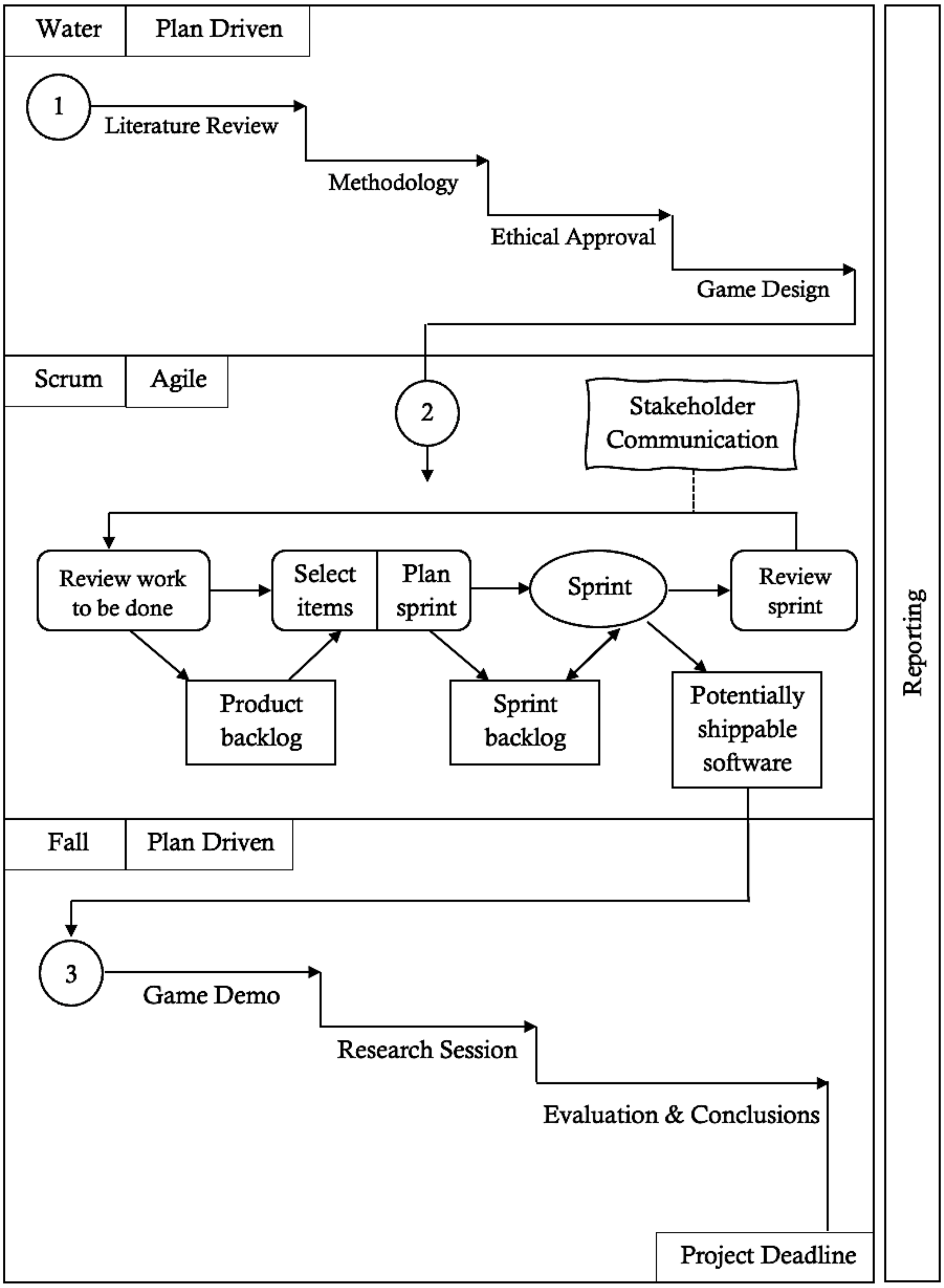


Figure 9: Water-Scrum-Fall

Figure 9 applies the Water-Scrum-Fall model to this project and its milestones. The Scrum sprint cycle (2) is adapted from Sommerville's *Software Engineering* (Sommerville, Ian, 2016, Figure 3.9, p. 86) with the additional 'Stakeholder Communication' activity introduced to reflect iterative client communication during the sprint phase.

3.4.1 Water Phase: (Weeks 1-8)

This phase establishes the foundational framework through sequential, documentation-intensive activities. Linking the literature review and methodology in the earlier weeks granted clarity for the ethical requirements needed. During the period waiting for formal approval, requirements for the game and it's design could begin, thinking about the game's content and mechanics, drawing inspiration from the literature review.

Following the formal approval, requirements were challenged alongside the User Centred Design, and stakeholder involvement began. Thereafter, the Game Design Document (GDD) functioned as a living document, evolving as design decisions emerged but providing initial direction for development. These phases inherently required waterfall-style sequencing because downstream activities depend critically on their completion.

3.4.2 Scrum Phase: (Weeks 9-19)

Development is organised into time-boxed sprints, each producing a tangible increment toward the evaluation-ready artifact:

- **Sprint 0** (1 week): Establish development environment, derive product backlog from GDD, configure Firebase deployment and version control.
- **Sprints 1-2** (2 weeks each): Implement core gameplay mechanics (hexagonal board state management, piece positioning, turn logic) and foundational UI (board rendering, piece interaction, game flow). These sprints prioritised stability, ensuring fundamental interactions function as intended.
- **Sprint 3** (2 weeks): Integrate Scottish Gaelic content with cultural authenticity, refine aesthetic elements and narrative framing, execute internal testing and refinement to achieve evaluation-readiness.
- **Sprint 4** (1-2 weeks): Include game 3, audio recordings.

This Scrum structure enabled me to detect and resolve technical issues incrementally rather than discovering them later. Each sprint produced a potentially shippable increment that could be tested internally, allowing design validation before the single-session evaluation. A 'standup' identified blockers, and sprint retrospectives captured lessons learned for subsequent iterations, a practice well-established in learning design contexts.

3.4.3 Fall Phase: Weeks 19 ongoing

Evaluation and report writing revert to a structured, deadline-driven workflow with limited opportunity for iteration. Qualitative data collection occurs in a single in-school session with the target user group, followed by thematic analysis of observations and post-test questionnaires. Once this evaluation session concludes, no further prototype iteration is feasible due to ethical constraints and the project timeline. The reporting phase proceeds linearly: data analysis, findings synthesis, discussion, and completion operate under a fixed university submission deadline.

3.5 Data Collection Methods

In line with the convergent mixed-methods design, quantitative and qualitative data are collected concurrently during a single research session at the school site. The strands are then analysed independently and subsequently merged during interpretation. This structure allows:

- Quantitative data to provide standardised indicators of usability and performance
- Qualitative data to provide rich insight into engagement, cultural authenticity, and language use, which numerical measures alone cannot fully capture.

This approach is particularly appropriate for game-based learning in minority language contexts, where both observable behaviour and subjective experience are central to understanding the artefact's effectiveness.

In this design, quantitative and qualitative data are collected concurrently during the same research session, analysed independently, and then merged during the interpretation phase. This approach allows for a comprehensive analysis, where the qualitative findings (the 'why') can be used to explain and contextualise the quantitative results, which can contain information about user behaviour that they cannot directly explain.

3.6 Experiment Design

This section outlines the participants, setting, apparatus, and procedure of the evaluation session, together with the specific quantitative and qualitative methods employed.

3.6.1 Participants and setting (TBD)

The primary participant group comprises upper-primary GME students (P6-P7, ages 8-12) enrolled at Bun-sgoil Steòrnabhaigh (Stornoway Primary School).

The sample size is x. This group reflects the critical P6-P7 stage identified in Chapter 2 as a key juncture for Gaelic proficiency and continued engagement.

A secondary participant group consists of GME teaching staff supervising the evaluation session **n =** . Teachers provide expert professional perspectives on cultural appropriateness, engagement, and classroom fit via a post-session questionnaire and, if required, brief follow-up clarification.

The evaluation takes place in the pupils' usual school environment, using school devices, to maximise ecological validity and align with the contextual emphasis of ISO 9241-210.

3.6.2 Mixed-methods rationale

The evaluation phase employs a mixed-methods design that integrates:

- Quantitative usability metrics, capturing effectiveness and efficiency of interaction; and
- Qualitative experiential insights, capturing pupils' and teachers' perspectives on engagement, cultural authenticity, and language use.

This design aligns with Creswell and Plano Clark's framework for mixed-methods research, in which quantitative and qualitative strands are collected in parallel, analysed separately, and then integrated during interpretation to produce a richer understanding than either approach alone. It also responds directly to Chapter 2's emphasis on combining behavioural evidence with culturally and linguistically situated accounts.

Methodological rationale

Game-based learning evaluation in this context requires attention to multiple dimensions:

- whether children can effectively use the system (usability);
- whether they find it engaging and enjoyable (user experience);
- whether cultural representations feel appropriate and authentic (cultural authenticity); and
- whether the game supports natural Gaelic language use, particularly idiomatic expressions (linguistic effectiveness).

Quantitative metrics offer comparability and clarity on performance and perceived usability, while qualitative methods capture nuances of cultural fit, identity, and language practice that are central to digital normalisation efforts.

Quantitative Methods

Table 6: Quantitative Methods

Method	Data Collected	Purpose	Alignment with ISO 9241-11
Adapted System Usability Scale (emoji Likert scale)	Perceived usability ratings across 10 standardised statements	Age-appropriate assessment of overall usability perception	Satisfaction
Screen recording	Interaction patterns, navigation pathways, error frequency, recovery strategies	Post-hoc analysis of how pupils navigate and problem-solve	Efficiency
Completion rate	Proportion of pupils successfully completing core gameplay tasks	Indicator of goal achievement within session duration	Effectiveness
Timestamps and duration analysis	Logged events (level completion, help access, pauses)	Analysis of pacing, difficulty calibration, and sustained engagement	Effectiveness & Engagement
Help button usage frequency	Number of help system accesses per pupil	Indicator of self-sufficiency and adequacy of in-game support	Efficiency & Independence

TBD - Note that I will be quantifying engagement/enjoyment as well as the cultural relevance / pride levels too

Qualitative Methods

Table 7: Qualitative Methods

Method	Data Collected	Key Focus Areas	Rationale
Pupil Discussion Group	Collective reflection through peer discussion. Thematic analysis of transcript.	Enjoyment, cultural authenticity, Gaelic language use during play.	Peer interaction reduces power imbalance stimulates richer discussion
Observational Field Notes	Structured documentation during session	Usability issues, engagement indicators, spontaneous Gaelic use, peer interactions	Capturing whether authentic language normalisation is gained, and there must be a way to record data throughout the session that could affect the validity of the study.
Teacher questionnaire	Expert professional judgement, from contextually grounded observations	Cultural authenticity of language/references, as well as observed pupil engagement and language use	Provides expert validation, and further promoting indigenousminority-language design ethics

3.6.5 Skill transfer, counterbalancing, and fatigue - remove

The study employs a single artefact rather than multiple experimental conditions; thus, formal counterbalancing is not required. Nonetheless, design choices aim to minimise:

- **Fatigue effects,** by constraining total session length to within a standard lesson and structuring activities (gameplay, questionnaire, discussion) to maintain variety.
- **Learning effects?**

Section 3.7 Ethical Considerations

Ethical approval for this study was obtained from the University of St Andrews Teaching and Research Ethics Committee, alongside formal permissions from the local authority and the participating primary school, before any recruitment or data collection. As the research involved children aged 8-12 in a school setting, particular attention was paid to safeguarding, informed consent, and data protection. Parent/guardian and teaching staff were provided with detailed participant information sheets and consent forms in advance, and only pupils with written parental consent and teacher consent were invited to take part. Pupils then gave verbal assent on the day of the session after the child's information script was read aloud. During the visit, the game-play session, pupil questionnaire, teacher questionnaire, and group discussion were all conducted under teacher supervision, with participants reminded of their right to stop at any time and to withdraw their data within a defined window after the visit. All data were pseudonymised, stored on secure university systems, and handled in line with GDPR and institutional policies, with consent forms kept separate from research data and identifiers destroyed after the withdrawal period. A full account of the ethical considerations, approvals, and data governance procedures is provided in Chapter 6.

Chapter 4 Game Design

Am fear nach cuir a shnaim, caillidh e 'chiad ghreim. *Air a chruinneachadh le Alasdair MacNeacail*	He that doesn't knot his thread will lose his first stitch. *Collected by Alexander Nicolson*

This chapter translates the theoretical and methodological foundations established in Chapters 2 and 3 into a concrete design for the Gaelic educational game. The first part (Section 4.1) follows the user-centred design principles set out in ISO 9241-210, focusing on understanding users, contexts of use, and culturally situated requirements through a user group profile, personas, and reflexive design decisions. This provides a structured account of how the needs of island GME pupils and teachers are elicited and transformed into design constraints. The second part (Sections 4.2-4.7) presents the Game Design Document (GDD), a structured blueprint covering the game's concept, mechanics, narrative, audiovisual direction, technical implementation, and accessibility, ensuring that the final artefact remains traceable to the user-centred rationale articulated earlier in the chapter.

4.1 UCD

Table : Examples of outputs from human-centred design activities UCD

Activities	Outputs from human-centred design	Examples of information contained in outputs
Understand and specify the context of use	Context of use description	— User group profiles — As-is scenarios — Personas
Specify the user requirements	User needs description User requirements specification	— Identified user needs — Derived user requirements — Required design guidance
Produce design solutions to meet these requirements	User-system interaction specification User interface specification Implemented user interface	— Scenarios of use — Low-fidelity prototypes — High-fidelity prototypes
Evaluate the designs against requirements	Evaluation results Conformance test results Long-term monitoring results	— Usability-test report — Field report — User survey report
NOTE More detailed information on each output can be found in ISO/IEC TR 25060.		

('Ergonomics of human-system interaction - Part 210: Human-centred design for interactive systems', 2019)

This project adapts the UCD framework to accommodate two significant constraints: (1) a single scheduled classroom visit for formal user evaluation, and (2) limited ongoing access to GME student participants for iterative testing. In this section, the understanding of the user group is developed further to ensure the game developed is built on top of the clearest understanding of the community as possible.

4.1.1 User Group Profile

Whilst the Gaelic-medium education (GME) cohort spans geographically and demographically disparate rural and urban contexts across Scotland, this investigation deliberately circumscribes its user group profile to pupils within a small island community in the Western Isles. This delimitation is a principled methodological choice: rather than generating a generic upper-primary profile applicable to all GME learners, the profile is constructed with sufficient specificity to capture the island-situated linguistic, cultural, and technological particularities that meaningfully differ from mainland and urban GME provision.

Role of the User Group Profile in Design

User group profiles, in HCI design practice, serve as repositories of key user characteristics and contextual affordances Preece et al., 2015) - TBD need to insert exact page numbers from this texbook.

They function as generative artefacts that inform subsequent design specifications, including personas, usage scenarios, and task models, ensuring that design decisions remain epistemically grounded in empirical user understanding rather than designer assumptions.

From User Characteristics to Design Constraints

In this project, the user group profile synthesises information about learners' age range, linguistic background, schooling experiences, and everyday Gaelic practices, together with the technological, curricular, and cultural context of a GME classroom in the Western Isles. This operationalises the critical insights from Chapter 2, which identified both the gap in idiomatic Gaelic transmission within formal GME settings (Section 2.1.5) and the necessity for "normative situations" that reflect authentic community linguistic and cultural practices (Section 2.2.3). The profile, therefore, moves beyond generic upper-primary characteristics to capture island-specific dimensions: pupils' home landscape and particular linguistic practices of island communities that differ from urban and mainland rural contexts. This in turn will inform a more tailored solution to those in this community.

This profile operationalises two critical insights from the literature review (Chapter 2). First, it directly addresses the documented attrition in idiomatic Gaelic transmission within formal educational contexts (Section 2.1.5). Second, it instantiates Veresov and Veraksa's (2023) conceptual framework of "normative situations" (Section 2.2.3), wherein game environments that embody culturally specific patterns of interaction, spatial contexts, and linguistic registers afford children internalisation of both language systems and culture-bearing social norms. By anchoring design to island-specific phenomena, distinctive settlement patterns and place-based linguistic indexicality, the profile ensures that the game functions as a space for cultural-linguistic immersion rather than decontextualised vocabulary rehearsal.

Ethnic Diversity and Inclusive Representation in GME

An important consideration emerges from demographic realities within the contemporary Gaelic community. Chapter 2 revealed the geographical spread of the GME community, however, ethnicity was not a well reseacredh area within GME, this required a bt more research, which found that the wider Gaelic population is increasingly ethnically diverse, with documented instances of learners from migrant and diaspora backgrounds achieving proficiency in Gaelic (*Stornoway Gazette*, 2021), (BBC 'Afro-Gaidheil', 2021).

This necessitates critical reflection on human representation within game design, since choices regarding character ethnicity, visual anthropomorphism, and identity-bearing imagery carry implicit inclusivity implications. As discussed in Section 2.2.5 on user-centred design for minority languages, Felicya and Lestari's Indonesian case study demonstrates how non-human avatars can support identification while reinforcing local cultural reference points: "The avatars, which represent students in the app, are chosen from Indonesia's exotic animals"(Felicya and Lestari, 2024, p. 5). Their findings suggest that player identification can operate robustly through non-human avatars, particularly when grounded in locally familiar fauna, which is advantageous in ethnically heterogeneous user populations because it circumvents representational essentialisms that risk marginalising pupils whose identities diverge from dominant visual framings of "Gaelic-ness".

By analogy, selecting animal avatars that resonate with the ecological and cultural landscape of island GME communities can both normalise Gaelic in everyday, locally meaningful situations and support inclusive identification across the full demographic spectrum of learners. This design consideration aligns with the broader user-centred design approach outlined in Section 2.2.5, ensuring that visual and narrative choices remain accountable to the lived diversity of the contemporary Gaelic community rather than to a narrow ethnic stereotype of the "Gaelic speaker".

4.1.2 Personas

Based on the User Group Profile, personas have been generated to guardrail important aspects of the game's design that should be considered in the Game Design Document (GDD).

Personas are fictional yet evidence-based archetypes that represent key user segments within a target population, designed to embody the goals, motivations, pain points, and behavioural patterns of real users. Developed and refined across product development cycles at Microsoft, personas function as a powerful communication mechanism that brings a shared understanding of users to entire teams, from designers and developers to testers and marketers (Pruitt and Grudin, 2003, p. 1).

As Pruitt and Grudin note, "Personas invoke this powerful human capability" - our innate ability to predict and extrapolate behaviour from partial knowledge of a person's mental state, a cognitive faculty grounded in Theory of Mind (Pruitt and Grudin, 2003, p. 2).

In this project, the three pupil personas (Isla, Tariq, and Tormod) and the teacher persona (Mrs Ross) synthesise empirical findings from the user group profile, literature on GME demographics and idiomatic Gaelic erosion, and HCI research on minority-language design discussed in Chapter 2. Together, they operationalise key design requirements - idiomatic language gaps, digital normalisation, normative cultural situations, and mixed linguistic proficiency - into concrete design constraints that inform the game's mechanics, narrative, linguistic register, and interface scaffolding.

PERSONAS TBD: Include links, also back up claims with Nicleoid's scripts

Isla Smith

Upper Primary GME Student

Demographics

- **Age**: 10 years old (Primary 6)
- **Location**: Outer Hebrides
- **Home Language**: Primarily English. Parents are non-Gaelic speakers
- **Technical Proficiency**: High - digital native, comfortable with tablets, smartphones, and web-based games
- **Gaelic Proficiency**: Fluent in classroom contexts. Limited idiomatic/colloquial exposure or casual Gaelic usage.

Quote

"I like speaking Gaelic at school, but there's nothing fun to do in Gaelic online. Everything's either a translation or for people learning Gaelic. I wish there was something cool that felt... actually Gaelic"

Background & Context

Isla has been in Gaelic Medium Education since Primary 1. Her parents enrolled her for heritage reasons (her great-grandmother spoke Gaelic) and the cognitive benefits of bilingualism. She's confident speaking Gaelic with teachers and classmates during school hours, but switches to English immediately after school.

Goals & Motivations

- Wants to feel connected to Scottish culture in a contemporary, non-museum way.
- Seeks fun, engaging digital experiences in Gaelic that feel as exciting as English games.
- Desires peer validation - wants to share cool Gaelic content with friends and understand Gaelic in the same depth as those with Gaelic speaking parents at home.
- Wants to understand the "real Gaelic" her elderly neighbour sometimes uses at community events.

Behavioural Patterns

- Plays mobile games daily (Roblox, Minecraft).
- Uses technology independently without adult supervision.
- Prefers short, rewarding gameplay loops over long narrative experiences.
- Engages most when actively participating rather than passively consuming.

Design Needs

- Challenge-feel game
- Self-guided gameplay requiring minimal adult support, parents are unable to explain idiomatic language.

Pain Points & Frustrations

- Most Gaelic digital content is educational rather than genuinely fun.
- Limited exposure to authentic idiomatic expressions that native speakers use.
- Demotivated to carry on learning a language she cannot interact with out with school.

Figure 10: Persona 1

EDIT - insert link

Tariq (طارق)

Upper Primary GME Student

Demographics

- **Age**: 10 years old (Primary 6)
- **Location**: Outer Hebrides
- **Home Language**: Arabic (Native), Parents have some English
- **Technical Proficiency**: High - digital native, comfortable with tablets, smartphones, and web-based games
- **Gaelic Proficiency**: Growing confidence in classroom contexts. No Gaelic exposure outside of school.

Quote

" I speak Arabic at home, Gaelic at school, and English everywhere else. My friends and I think it's cool I know so many languages. However, I don't need a game to teach me about being Scottish - I AM Scottish. Gaelic is part of my life now, not something I'm visiting. I'm not just watching it from outside. I hope digital advancements in Gaelic capture this"

Background & Context

Tariq's family resettled in the Outer Hebrides when he was 6 years old. Unlike families who choose GME for heritage or ideological reasons, Tariq's family had no prior connection to Gaelic culture. GME is now the default option for education provision for children in the Western Isles[INSERT LINK]. Tariq was keen to to learn the local language from the beginning, and after discovering the benefits of GME, his parents considered it a great option. However, Tariq finds it challenging to practice Gaelic and relies solely on school for this. Although he can bring Gaelic resources home, they aren't always engaging and never interactive.

Goals & Motivations

- Wants to feel connected to Scottish culture while maintaining Syrian identity.
- Motivated by the novelty of multilingualism
- Now past the learning phase, he wants to actively use the language in as many forms as possible
- Hopes to make parents proud despite their inability to speak Gaelic or help with homework

Behavioural Patterns

- Digital native across cultures: Consumes content in Arabic, English, Gaelic - navigates multiple cultural digital spaces
- Rapid language acquisition: Learn as much as possible but may not understand the true meaning behind words or phrases

Design Needs

- Multicultural Identity Representation (content that doesn't assume monocultural Scottish heritage)
- Linguistic Accessibility - Self-guided tutorials essential, having Gaelic idioms explained

Pain Points & Frustrations

- Triple language juggling: Must maintain Arabic at home, master English for wider Scotland, excel in Gaelic for school
- Parental language gap: Parents cannot support Gaelic schoolwork due to language barrier
- Interrupted education: Missed some early schooling years, playing "catch-up" means more independence and reliance on others for explaining local concepts they seem to 'just know already'

Figure 11: Persona 2

Tormod Moireach

Upper Primary GME Student

Demographics

- **Age**: 11 years old (Primary 7)
- **Location**: Isle of Lewis, Western Isles
- **Home Language**: traditional Gaelic-speaking family: parents, grandparents, extended family all fluent native speakers
- **Technical Proficiency**: High - digital native, comfortable with tablets, smartphones, and web-based games
- **Gaelic Proficiency**: Growing confidence in classroom contexts. No Gaelic exposure

Quote

"I speak Gaelic very well, probably better than English as I've spoken it my whole life. But I don't like most of the Gaelic-focused things in my community, like dancing and music, I prefer gaming and football. I'd like to use more of my traditional Gaelic; it's pretty funny sometimes and super relatable, but literally nobody under 40 talks like that. If I started speaking my home Gaelic at school, people wouldn't understand it or think I spend too much time with older people..."

Goals & Motivations

- Reduce the stereotype of 'traditional Gaelic phrases is only for old people'
- Peer Understanding - If a critical mass in his class can relate traditional phrases to a game rather than associate them with an older community the ability to use such Gaelic casually is increased.
- Increase the amount of casually used Gaelic, allowing authentic use, not forced in spaces that are appealing to him

Background & Context

Tormod is a native heritage speaker of Scottish Gaelic, where idiomatic Gaelic weaves naturally into conversations at home. His diversifying peer group make English the norm, his entire digital world is exclusively English and he became aware that speaking Gaelic is uncommon and abnormal for teenagers outside his island. Previously, Tormod used traditional Gaelic sayings but his friends didn't understand them and made him feel that his family's use of seanfhacail is "old-fashioned" and embarrassing, triggering a response that drives adolescent abandonment of heritage language.

Behavioural Patterns

- Never uses Gaelic socially with any peers
- Enjoys short gaming app sessions that are non-academic and modern
- Enjoys fishing and crofting
- Replies to family in English instead of Gaelic as his friends do not speak Gaelic to their parents at home

Design Needs

- Reduce the educational framing of idioms, don't separate them from casual use
- Contemporary, Not overly 'heritage' focused design
- No punishment for errors

Pain Points & Frustrations

- Digital Disconnect - Fluent in Gaelic with zero cool applications; feels useless in domains he values
- Peer Conformity Pressure – Speaking Gaelic in a traditional way isn't 'cool' or understood by peers
- Linguistic Insecurity: Hears GME learners' different Gaelic; questions if his "traditional" variety is legitimate

Figure 12: Persona 3

Mrs Ross

Primary 5-7 GME Teacher

Demographics

- **Age**: 45 years old
- **Position**: Primary 5-7 GME teacher (15 year's experience)
- **Location:** Rural GME school, Western Isles
- **Gaelic Background:** Native speaker, grew up in Gaelic-speaking household
- **Technical Proficiency**: High - uses, technology often for work and at home
- **Qualifications:** PGDE (Gaelic Medium), MA in Gaelic Studies

Quote

"I can teach them the language, but I can't teach them to live it. My teaching now has stark switches between local tradition and global innovation, and joining the two is becoming harder to do. Gaelic doesn't need to be preserved like a museum piece; it should feel normal - not old-fashioned, not just translated - but genuinely part of their lives, which are increasingly digital."

Background & Context

Mrs Ross is a passionate GME practitioner who has seen the sector grow from small composite classes to a recognised mainstream option in her authority. Many of her current pupils come from non-Gaelic-speaking homes, and for them the classroom is the primary - sometimes only - domain where Gaelic is used. She recognises that while her pupils can meet curricular targets, their exposure to idiomatic and colloquial Gaelic, including seanfhacail, is minimal compared to what she experienced growing up. She is acutely aware of the tension between institutional provision and community transmission: pupils achieve formal proficiency but rarely transfer Gaelic into their home, peer, or online lives.

Goals & Motivations

- **Ethnolinguistic identity, not just attainment:** Wants pupils to see Gaelic as "their" language - something that belongs in jokes, games, and everyday chat, not just in reading passages or formal writing.
- **Digital normalisation:** Seeks resources that normalise Gaelic in contemporary contexts
- **Bridging diverse profiles:** Desires to minimize achievement gap between heritage speakers and learners
- **Idiomatic integration:** Aims to incorporate seanfhacail into curriculum lacks engaging methods

Behavioural Patterns

- Dedicates 3-4 hours weekly to lesson preparation and marking beyond contracted time
- Avoids using devices for pupils in the classroom freely, as they usually result in conformity to English resources

Design Needs

- **Culturally authentic content** incorporating idiomatic expressions and normative situations
- **Minimal teacher supervision** required - self-guided tutorials and feedback systems
- **Ability to track pupil engagement** (optional analytics)

Pain Points & Frustrations

- **Translation over design:** Most digital resources are translated, not culturally designed
- **Poor fit of existing Gaelic games:** Existing Gaelic games are either poorly localised content or basic vocabulary drills
- **Mixed proficiency in one class:** Time-consuming to create culturally authentic materials from scratch

Figure 13: Persona 4

4.1.3 Reflexive Design Practice

My unique position as an insider researcher is methodologically valuable. Growing up in the Outer Hebrides, experiencing both institutional GME education and community transmission gives me what (Muhammad *et al.*, 2015 p. 1046) calls "matched identities" with my research participants. This insider status facilitates trust, access, and ecological validity of knowledge production.

As a local of the area and a former GME pupil, I possess situated, experiential knowledge of this environment. This insider perspective is used reflexively, rather than anecdotally, to sensitise design decisions to idiomatic expressions, culturally indexical elements and the informal language practices that characterise authentic Gaelic use in family and peer contexts. This approach aligns with the indigenous design ethos of "nothing about us, without us" (Section 2.2.4), ensuring that cultural representation is grounded in lived community knowledge rather than external stereotypes.

My lived experiences assists with my ability to distinguish between indexical and iconic elements to ensure resonance. My knowledge of which seanfhacail are age-appropriate and the contexts where idioms are used organically addresses NicLeòid's finding that most GME pupils have little exposure to idiomatic Gaelic unless there's community input.

4.1.4 Designing Game Content

In order to achieve localisation, I bring together indexical and iconic elements rooted in the Outer Hebrides and collect Gaelic idioms, both ocean-themed and more general, for use throughout gameplay.

Game Setting: Beach & the Coast

The location to which I will be carrying out my game's evaluation is on the Isle of Lewis, therefore I have chosen to localise towards the landscape. As an island, beaches, rocks, seals, and fishing boats are prominent features. Similarly, children of all linguistic and cultural backgrounds living in Hebridean communities grow up surrounded by marine life and coastal ecology as normative features of their environment.

Designing Indexical Elements through Idiomatic Gaelic

When reflecting on normative transmission of idiomatic Gaelic, I observed distinct categories of cultural knowledge in archives, dictionaries, and other literature, and separate into indexical and iconic elements:

Pragmatic Idioms

(Fraser, 1999, p. 943) clearly defines interjections such as "Oh!" "Wow!" and "Shucks!", noting that "they must be treated as pragmatic idioms". Such non-context-dependent linguistic markers exist in Scottish Gaelic. Words such as *"nise", "och",* and *"duda?"* "*Obh obh*" are versatile pragmatic idioms. Like "Oh!" "Wow!" and "Shucks!", they may not be taught formally, but rather organically reproduced through colloquial exposure.

Table 8: Pragmatic Idioms

	Phrase / Saying	**Translation / Source**
Pragmatic Idioms	och	1) sigh 2) oh!
		('och', Faclair na Gàidhlig / The Gaelic Dictionary)
	nise	Now - discourse marker, as defined by (Fraser, 1999)
		Used often in audio recording from archives: (MacIntyre, Donald, 1899-1964, 1962) & 09/02/2026 16:49:00 ('a-nise', Faclair na Gàidhlig / The Gaelic Dictionary)
	duda?	huh? eh? whussat? (in the sense of sorry, excuse me, pardon?)
		('duda', LearnGaelic Dictionary)
	obh obh	"Oh Oh"
		Recorded at time 15:45 in recording from archives: (MacDonald, Duncan (1882-1954), 1953)
	Seadh	"yes, uh huh!"
		(seadh, LearnGaelic Dictionary) (MacIntyre, Donald, 1899-1964, 1962)
	Aidh aidh	aye aye
		(MacIntyre, Donald, 1899-1964, 1962)
	Haoidh	"hey! ho!"
		(haoidh, Faclair na Gàidhlig / The Gaelic Dictionary)

Context Specific Idioms

Lamb notes that linguistic conventionalisation (the normative adoption of a word or phrase by a community of language users) is more likely when expressions are frequent, socially salient, and memorably encoded through features such as strong imagery, rhythm, alliteration and assonance (Lamb, 2015, p. 233, drawing on Coulmas, 1979 and Ong, 2005).

Table 9: Context Specific Idioms

		Phrase / Saying	Translation / Source	Game Design Comment
Context Specific Idioms	Oceanic Idioms	*"An giomach, an rionnach is an ròn, trì seòid a' mhara"*	The lobster, herring and the seal: the three jewels of the sea. (‘giomach, rionnach is ròn, trì seòid a’ chuain’, Faclair na Gàidhlig / The Gaelic Dictionary)	This proverb names three marine creatures, but it does more than list them: The three-part structure incorporating rhythm alliteration and assonance makes the saying memorable and easy to reproduce orally. Integrating a lobster, seal and herring as distinctive game characters can reinforce the idiom with imagery and through repeated use in play.
		"Tha iad cho coltach ris an dà sgadan"	“They are as alike as two herring” English equivalent could be “like two peas in a pod, very similar to each other” (LearnGaelic Dictionary)	Drawing on this idiom within a game context, a context-specific gamified task could require learners to identify, pair, or match two highly similar items (for example, visually similar characters or objects that form a pair), so that successful completion of the task is tightly coupled to understanding and using the phrase in an appropriate semantic and situational frame.
		“San Earrach, nuair a bhios a chaora caol, bidh am maorach reamhar.”	“In Spring, when the sheep are lean, The shell-fish will be fat” (MacDonald, T. D., 1926, No. 510)	The proverb's seasonal reference and maritime imagery are particularly well-suited to the game's oceanic theme and align with the artifact's planned evaluation in Spring 2026.
	Reward Idioms	*“Nach Buidhe dhut”*	“Aren’t you lucky?” (‘Nach buidhe dhut’, LearnGaelic Dictionary)	Gaining a point or achievement within the game could prompt integration of these proverbs.
		“Sin thu fhèin”	There you are! (also Well done! Good for you!) (SpeakGaelic, 2024)	

Physical Indexical Elements

Tweed

As Harper and McDougall note, the Harris Tweed Act defines the cloth as "a tweed handwoven by the islanders at their homes and finished in the islands of Harris, Lewis, North and South Uist, Benbecula, and Barra (The Outer Hebrides), and made from pure virgin wool dyed and spun in the Outer Hebrides" (Harper and McDougall, 2012, p. 78). For that reason, tweed could function as an indexical element that acts as a visual anchor of local identity, unlike other indexical elements requiring linguistic knowledge or formal explanation or ceilidh performances, which demand embodied participation.

The Cairn

"In Gaelic, the word carn or cairn refers equally to natural and cultural landscape features, meaning both a 'rocky hill or mountain' and a 'heap of stones' (MacLennan 1979: 73). As a cultural form, cairns have been raised up in Scotland as boundary markers, way markers, summit markers and, notably, as grave markers, burial places and memorials for millennia. Formed from the most durable of materials, they seem to have the ability to fix memories in stone and in place." (Basu, 2022, p. 118)

The cairn's significance could be further reinforced through the seanfhacal "Nithear càrn mòr bho chlachan bheaga" ("A big cairn is built from small stones"), which functionally encodes the principle of incremental progress and sustained effort apt framing for the accrual of points and achievements within the game.

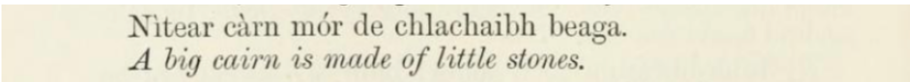

Nìtear càrn mór de chlachaibh beaga.
A big cairn is made of little stones.

Figure 14: Exploring the archives

From "A collection of Gaelic proverbs and familiar phrases" (Nicolson, Alexander & Macintosh, Rev. Donald, 1882, p. 334)

Local Animals as iconic elements:

The fish and animals around the Hebridean coast all have their own Gaelic names, from shore dwellers to deep-sea creatures. Incorporating these into the game creates familiar characters whilst providing learning opportunities for Gaelic vocabulary. The seal (ròn) was selected as the lead character. The name 'Ruairidh' was chosen for the seal. By using alliteration: "Ruairidh a' Ròn", it makes the lead character phonetically memorable and easy for children to say. Fish were initially considered but rejected because they could be confused with other game elements. A seal, being visually and behaviourally distinct, serves as a clear guide character to assist children through gameplay.

4.2 Game Design Document (GDD)

Executive Summary

This game is a browser-based game, designed to support Gaelic language usage and digital normalisation among P6-P7 students (ages 9-12) in Scottish Gaelic Medium Education (GME) settings. The game immerses players in an exclusively Scottish Gaelic linguistic environment through three distinct gameplay mechanics that reinforce vocabulary, idiomatic language patterns, and cultural authenticity specific to Hebridean maritime contexts.

Game Overview

The game features a seal character, Ruairidh an Ròn (Ruairidh the Seal), who guides players through a narrative framed around traditional Gaelic maritime culture. Players progress through three sequential games:

Game 1: Glac an Giomach (Trap the Lobster): A spatial-logic puzzle requiring players to strategically position stones on a hexagonal grid to encircle an autonomously moving lobster before it reaches the board perimeter. Players earn 1 point per lobster trapped within a 4-minute time constraint.

Game 2: Cho coltach ris an dà sgadan (As alike as two herring): A memory-based card-matching exercise featuring 12 cards with objects discovered at sea underneath. Players match pairs through recall, gaining 1 point for each correct match. Time taken up to 4 minutes.

Game 3: Cho luath ris a bhradan! (As fast as a salmon) A reaction-based identification task requiring players to capture specified fish species within a 3-minute time. The game implements progressive difficulty through three depth zones, each introducing distinct sets of local species. Points differ for different fish types, and a combo-based scoring mechanism rewards consecutive correct identification. Environmental cleanup is incentivised through bonus points for collecting ocean debris.

Active gameplay for approximately 12-15 minutes. Scoring is balanced across games. An initial game settings tutorial will guide children through the game buttons. There will be a tutorial for each game which the user will need to progress though before starting.

Design Philosophy

The game prioritises:

- Gaelic as default mode: 100% Gaelic linguistic presentation with no English fallbacks, establishing Gaelic as the normative interaction language. (The game has been Gaelic-first developed)
- Self-Guided Autonomy: Minimising adult supervision through intuitive UI and integrated tutorial systems
- Cultural Authenticity: Authentic Gaelic maritime contexts and idiomatic expressions drawn from traditional sayings (seanfhacail)
- Technical Accessibility: Cross-platform browser compatibility (HTML5/CSS3/JavaScript) suitable for school technology infrastructures
- Engagement: Immediate feedback, clear progression, and challenge pacing to sustain motivation over a short session.

Design Requirements

User Needs Description

To better understand user needs a meeting was held with teaching staff to discuss their thoughts on the games and their technical setup.

Script to be included in Appendix

User Requirements Specfication

Functional Requirements
FR1 - Gaelic Language Integration: All gameplay text, audio, instructions, and interface elements shall be presented in Scottish Gaelic, with no English fallbacks, to establish Gaelic as the default mode of interaction. **FR2 - Cultural Authenticity:** Game scenarios, characters, and narrative elements shall reflect authentic Gaelic cultural contexts through normative situations familiar to GME students. **FR3 - Idiomatic Language Use:** Gameplay shall incorporate age-appropriate seanfhacail (traditional sayings) and idiomatic expressions as natural game elements, supporting transmission of colloquial Gaelic. **FR4 - Self-Guided Play:** In-game tutorials and feedback systems shall enable autonomous gameplay without requiring adult assistance, meeting the secondary objective of minimising supervisor support needs. **FR5 - Progress Feedback:** The system shall provide clear, immediate feedback on player actions to support learning and engagement. **FR6 - Age-Appropriate Interaction:** Interface interactions shall suit P6-P7 developmental capabilities, including motor skills, reading comprehension levels, and attention span considerations.

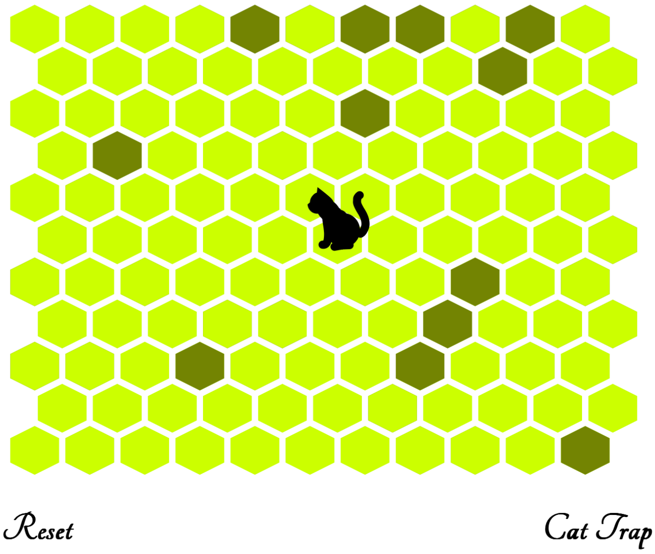
Non-Functional Requirements
Non-Functional Requirements specify how the system performs: **NFR1 - Platform Compatibility:** The game shall run on standard school-provided tablets/laptops via modern web browsers without requiring software installation, ensuring accessibility within GME school technology infrastructures. **NFR2 - Performance:** Initial load time shall not exceed 5 seconds; in-game response to user input shall occur within 0.5s. - measure this? **NFR3 - Session Duration:** Active gameplay shall stay within a 10-20 minute window. **NFR4 - Accessibility** (Potential) **NFR5 - Data Collection:** The system shall log anonymised interaction data (timestamps, completion rates, help button usage) without collecting personally identifiable information. **NF6 - potential - Offline Resilience:** Core functionality shall operate without continuous internet connectivity after initial load, accommodating potential school network limitations.

Game Plans:

Game 1: Glac an Gìomach (Trap the Lobster)

Design Rationale:

Game 1 will use spatial-logic puzzle mechanics to engage players in strategic planning whilst dropping conversational Gaelic vocabulary. The hexagonal grid system was selected over traditional square grids to increase spatial complexity and provide more nuanced pathfinding challenges, aligning with Wang's (2023) recommendation for "procedural variation" to sustain engagement beyond initial play (Section 2.3.2). The lobster should communicate un Gaelic, using idiomatic Gaelic, which positions the game mechanic itself as a vehicle for idiomatic transmission, addressing the gap identified by NicLeòid (Section 2.1.5).
Inspiration:



https://trap-thecat.com

Three major design changes:

- When creating a wall around the cat, all spaces within the circle need to be filled, this seemed redundant and unexciting so I wanted to ensure the game knew when lobster was trapped in any way.
- The gameboard just filling in the hexagon a darker shade is boring, rocks shoud be palced instead, thus giving a realistic blockage to the lobster.
- When the cat gets the edge, you must place one final dark green hexagon before it runs away, this too seems redundant. I would ensure that the lobser once is on an edge piece, runs at the ifrst opportunity.

Time: Initially 5 minutes, after teachers played the game in demo, it was reduced to 4.

Core Concept

Players must place stone barriers on a hexagonal beach grid to trap a lobster before it escapes on one of the edges of the board. The lobster actively seeks the nearest escape route, creating a turn-based strategic contest between player and AI.

Player Experience Goals

Drawing from the MDA framework (Section 2.3.1), Game 1 targets the following aesthetic responses:

- **Challenge**: Strategic depth through predicting lobster movement and planning barrier placement
- **Discovery**: Learning the lobster's pathfinding behaviour and optimal trapping strategies
- **Sensation**: Satisfaction when a trap closes successfully

The game should feel like outsmarting a cunning opponent rather than completing a static puzzle, dynamic enough to meet Rodríguez et al.'s (2020) finding that children prefer "dynamic and interactive tasks" requiring substantial input (Section 2.3.3).

Basic Mechanics

Board Structure

- Hexagonal grid representing a sandy beach
- Lobster spawns near the centre
- Board edges represent escape routes
- Some tiles pre-blocked with rocks to create initial variety

Turn Flow

- Player clicks an empty sand tile to place a rock
- Lobster responds by moving one tile toward its preferred escape route
- Cycle repeats until resolution

Lobster Behaviour

The lobster should appear to "think", not randomly, but actively seeking freedom. It should take the most direct available path to any edge. When blocked, it should recalculate and try alternative routes. The specific pathfinding approach will need to balance feeling intelligent without becoming unbeatable.

Win/Lose Conditions

- **Win (Player)**: Lobster has no valid moves remaining (surrounded by rocks and/or board edges)
- **Lose (Player)**: Lobster reaches any edge of the board

Each outcome triggers a board reset, allowing multiple attempts within the session timer.

Gaelic Integration

The lobster serves as a vehicle for idiomatic exposure through speech bubbles displaying pragmatic idioms (Section 4.1.4, Table 8):

- Movement exclamations: "haoi!", "duda?", "obh obh"

- Reaction phrases when trapped or escaping

This passive integration follows the stakeholder feedback that Gaelic should be "an added extra" rather than obviously foregrounded (Meeting 2 Notes, Section x).

Scoring & Feedback

Points are visualised through the cairn metaphor established in Section 4.1.4, reinforcing the seanfhacal "Nithear càrn mòr bho chlachan bheaga" (A big cairn is built from small stones). Each successful catch adds a stone to the player's cairn, as a culturally grounded progress indicator.

Visual & Audio Direction

- **Setting**: Beach/coastal scene consistent with the Hebridean maritime theme (Section 4.1.4)
- **Tiles**: Golden sand hexagons with rock obstacles
- **Audio**: Ambient ocean waves and seagulls to establish the normative coastal situation (Section 2.2.3)

Open Questions for Implementation

- What grid dimensions balance challenge with achievability for P6-P7 players?
- How should difficulty vary across rounds (if at all)?
- What rock density creates appropriate initial challenge?
- How frequently should speech bubbles appear without becoming distracting?

Visual Design:

INSERT MY LOW FIDELITY SKETCH HERE - (Game 1)

Game 2: Cho Coltach ris an Dà Sgadan (As Alike as Two Herring)
Design Rationale Game 2 targets memory and recall through a card-matching activity, providing cognitive variety from Game 1's spatial reasoning. The game title itself is an idiomatic expression, "Tha iad cho coltach ris an dà sgadan" (They are as alike as two herring), making the core game activity a direct enactment of the idiom's meaning (Section 4.1.4, Table 9). This tight coupling between mechanic and idiom exemplifies the approach advocated in Section 2.2.2: creating digital play experiences where language acquisition occurs "naturally and meaningfully" through gameplay rather than through "decontextualised vocabulary drills." **Core Concept** Players flip cards to find matching pairs of maritime objects. Cards display familiar coastal items when revealed, reinforcing Gaelic vocabulary for local sea life and objects. **Player Experience Goals** Drawing from the MDA framework (Section 2.3.1): • **Challenge**: Recalling card positions across increasing numbers of revealed cards • **Discovery**: Learning Gaelic names for familiar coastal objects • **Sensation**: Satisfaction of successful recall and pair completion The pace should feel contemplative compared to Game 1, a "breather" that still requires active engagement, meeting FR6's requirement for age-appropriate interaction. **Basic Mechanics** **Card Grid** • Fixed grid of face-down cards • Each card has a matching pair somewhere on the grid • Cards display tweed patterns when face-down (indexical element, Section 4.1.4) **Turn Flow** • Player flips first card (reveals object) • Player flips second card • If match: both cards remain face-up, point awarded • If mismatch: brief viewing period, then both cards flip back **Memory Challenge** Players must track which objects they've seen and where, updating their mental model with each reveal. The challenge scales naturally as more cards are revealed and the remaining options narrow.

Win/Lose Conditions

- **Win**: All pairs successfully matched
- **Completion**: Game ends when all pairs found (no "lose" state, completion-focused rather than failure-focused)

Gaelic Integration

Card objects display their Gaelic names upon reveal, providing incidental vocabulary exposure for maritime items. Objects should be:

- Culturally authentic to Hebridean coastal contexts (Section 4.1.4)
- Visually distinct for clear matching
- Named in Gaelic that pupils may recognise or encounter locally

Scoring & Feedback

Successful matches contribute to the cumulative cairn score, maintaining visual consistency with Game 1. Additional feedback might include:

- Animation confirming matched pairs
- Ruairidh's verbal encouragement using reward idioms (Section 4.1.4, Table 9): "Sin thu fhèin!", "Nach buidhe dhut!"

Visual & Audio Direction

- **Card backs**: Harris Tweed patterns (indexical element grounding the game in local material culture)
- **Card faces**: Clear illustrations of maritime objects
- **Setting**: Consistent coastal visual theme
- **Audio**: Calmer ambient audio than Game 1; gentle feedback sounds for matches

Open Questions for Implementation

- How many card pairs balance the challenge with session time constraints?
- Should matched cards disappear or remain visible?
- What viewing duration for mismatched cards optimises the memory challenge?
- How might the game accommodate players who complete quickly versus those who need more time? - TBD extension if time

INSERT MY LOW FIDELITY SKETCH HERE - (Game 2)

Game 3: Cho Luath ris a' Bhradan (As Fast as a Salmon)
Design Rationale Game 3 introduces reaction-based mechanics requiring attention, visual search, and quick response, completing the cognitive diversity across the three games. The title idiom "Cho luath ris a' bhradan" (As fast as a salmon) frames speed and agility as the core challenge. This game emerged from stakeholder feedback requesting "some kind of reaction game" with "moving objects on the screen" (Meeting 2 Notes, Section 5). The fishing/catching theme maintains the maritime normative situation (Section 2.2.3) while introducing dynamic gameplay that contrasts with the more deliberate pace of Games 1 and 2. The game implements Wang's (2023) recommendation for "evolving challenge structures" through progressive depth zones, each introducing new species and increasing difficulty (Section 2.3.2). **Core Concept** Players must catch specific fish species as requested by Ruairidh, who calls out orders in Gaelic. Fish swim across the screen at varying speeds and depths. Players tap/click to catch the correct species while avoiding incorrect catches. **Player Experience Goals** Drawing from the MDA framework (Section 2.3.1): • **Challenge**: Identifying correct species under time pressure • **Sensation**: Fast-paced, arcade-style excitement • **Discovery**: Learning to distinguish local fish species and their Gaelic names This addresses Rodríguez et al.'s (2020) finding that children engage most with "dynamic and interactive tasks" (Section 2.3.3), providing the most action-oriented experience in the game suite. **Basic Mechanics** **Play Area** • Underwater scene with fish swimming across • Multiple depth zones representing different marine environments • Fish enter from edges and swim across at species-specific speeds **Catch Mechanic** • Ruairidh requests a specific species in Gaelic • Player must identify and tap/click the correct fish • Correct catch: points awarded, new request issued • Incorrect catch: penalty or feedback

Progressive Zones

The game should transition through depth zones over the session duration:

- Shallower waters: slower fish, more common species, slow pace
- Deeper waters: faster fish, rarer species, increased challenge

This progression maintains engagement across the play session and rewards developing familiarity with species identification.

Win/Lose Conditions

- **Session-based**: Game runs for fixed duration
- **Scoring**: Cumulative points based on successful catches
- **No hard failure**: Mistakes reduce score but don't end the game

This approach maintains engagement for all proficiency levels while rewarding skilled play.

Gaelic Integration

TBD

Environmental Element

An optional secondary mechanic could involve collecting ocean debris (plastic, rubbish) for bonus points, introducing environmental awareness while providing additional interaction targets during slower moments.

Scoring & Feedback

- Base points per correct catch, potentially varying by species rarity/difficulty
- Visual and audio feedback confirming catches
- Running score displayed; contributes to cumulative cairn total
- Potential combo/streak rewards for consecutive correct catches
-

Visual & Audio Direction

- **Setting**: Underwater Hebridean waters with appropriate marine life
- **Fish**: Distinct, recognisable illustrations of local species
- **Movement**: Smooth swimming animations, speed varies by species and depth
- **Audio**: Underwater ambience, more energetic music than Games 1-2, clear audio for Ruairidh's requests

Open Questions for Implementation

- How many fish species are learnable within a single session?
- What request frequency balances challenge with processing time?
- Should fish names appear visually or only aurally?
- How should zone transitions be signalled to players?
- What penalty (if any) for catching wrong fish?
- How might speech bubbles or labels work with moving targets?

INSERT MY LOW FIDELITY SKETCH HERE - (Game 1)

Chapter 5 - Implementation

Cuir do lùth ris an obair, agus ionnsaichidh an obair thu. *Tobar an Dualchais, Leòdhas*	Put your energy to the work, and the work will teach you. *Tobar an Dualchais, Lewis*

This chapter documents the technical implementation of the game artefact, with particular focus on novel algorithms, unusual design decisions, user interface innovations, and the iterative refinements that shaped the final product. The implementation followed the Water-Scrum-Fall methodology outlined in Section 3.4, with discovery and adaptation occurring throughout the development sprints.

5.1 Technical Architecture Overview

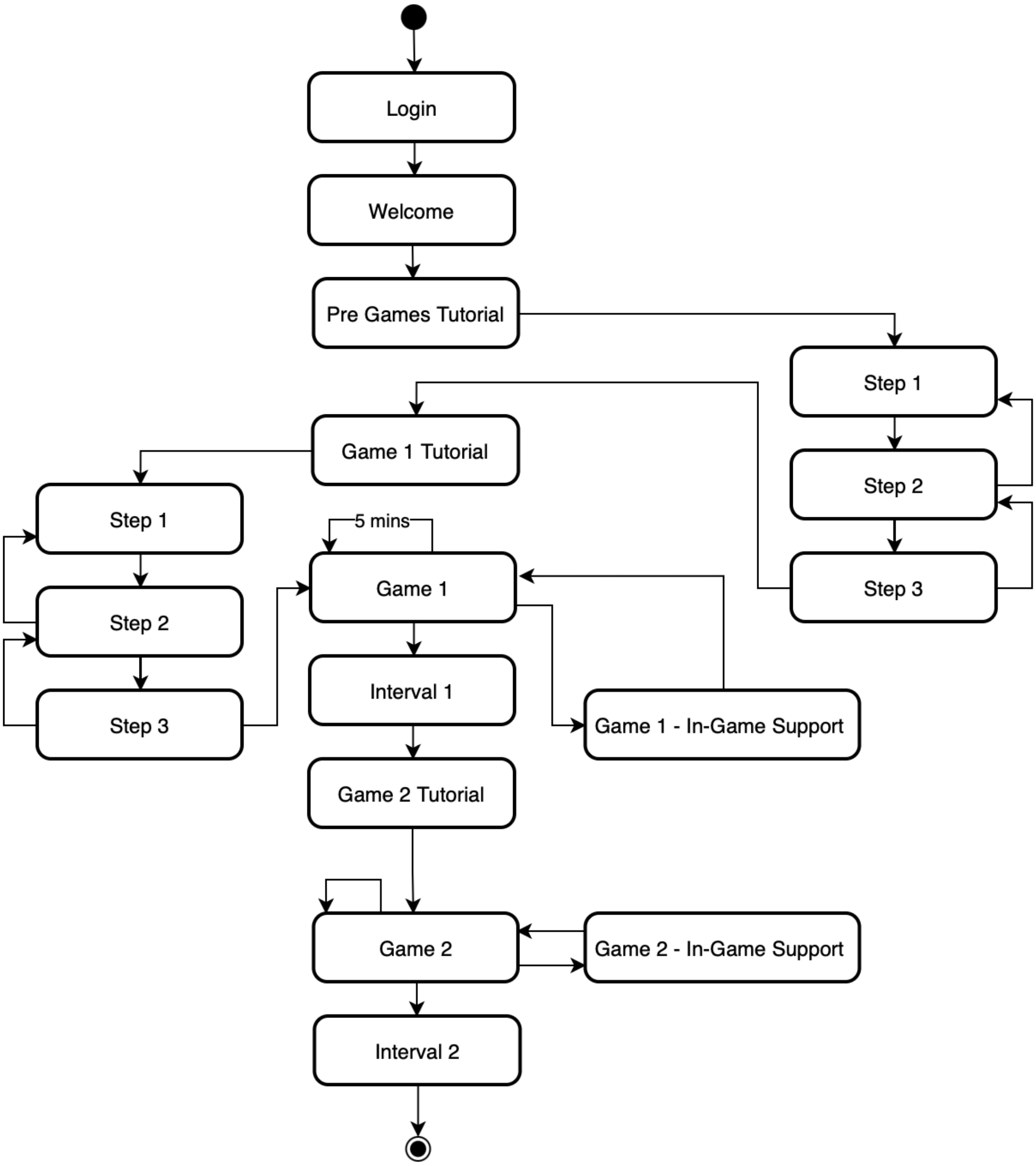
The game was implemented as a single-page web application using vanilla HTML5, CSS3, and JavaScript, deliberately avoiding external frameworks to ensure maximum compatibility with school Chromebooks and tablets (NFR1). This approach, while increasing initial development complexity, eliminated dependency risks and reduced bundle size to approximately X KB (excluding SVG assets).

The decision to consolidate all JavaScript into a single file, simplified deployment to Firebase and eliminated module-loading issues encountered during early testing on school networks with intermittent connectivity.

5.2 State Machine Architecture

The game flow is managed by a central GameFlowController class implementing a finite state machine pattern.

Game Mechanics and Systems:



TBD: Update to include Game 3

Writeup TBD:

Game 1: Hex Grid Pathfinding

Explain my string hash keys

BFS for lobster shortest escape path to edge

Rotation mapping for 6 directions

Card Matching

Sets for flipped/matched indices O(1) Lookup

Game 3:

Weighted random spawning

3 zones

Fish phsyics movements - shoaling, scurry etc.

Audio System

AudioManager using.ogg tracks - exlain why for looping

Tutorials & Help

Key Bugs Fixed

Iterations & Optimisations

Scrum Backlog

- **Include my calendar etc. - appendix?**

Chapter 6 - Ethics

Any ethical considerations for the project. Your initial ethics form, and full ethics 'favourable opinion' letter if applicable, should be included as an appendix.

TBD Include a short writeup, opinion letter and all 13 other forms to be included in the Appendix.

USER-CENTERED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION

Phases of Ethical Significance

Phase 1: Recruitment, Ethics & Preparation (Weeks 1-4) – School Open Monday 5th January
Goal: Secure access, verify safety, and obtain legal consent

Step	Operation / Action	Party Involved	Documents & Forms		Timeline / Duration
1.0	Ethics & Permission	Researcher Ethics Committee Supervisor	- UTREC Approval (Univ. St Andrews) - Local Authority Approval (Comhairle Nan Eilean Siar) - School Approval	DEC 2025 - JAN 2026	Pre-contact
1.1	Initial Contact	Researcher to School Head	- letter-to-school-in-principle.pdf		Email sent
1.2	Verification Meeting	Researcher & Head Teacher	- Agenda: Confirm dates, logistics, and eligible classes.		15–30 mins
1.3	Formal Confirmation	School to Researcher	- letter-to-school-commencing.pdf		Written confirmation
1.4	Teacher Mock Session	Researcher & Class Teachers	- Activity: Demo of game prototype. - Goal: Verify cultural/age appropriateness & safeguarding. - Outcome: Greenlight for pupil sessions.		30 mins
1.5	Consent Distribution	School Admin / Teachers	- PG_PIS pdf (Parent Info Sheet) - PG_Consent-form pdf - letter-to-parentguardian.pdf 5 x - T_PIS pdf (Teacher Info Sheet) - T_Consent Form pdf		15–20 mins (admin time)
1.6	Consent Collection	Parents &Teachers to School	- Signed Consent Forms (Must be returned to participate).		1 week window
1.7	Pupil Selection	Class Teacher	- Action: Cross-reference signed forms with eligible pupils (GME, 8-12y).		15–30 mins

Phase 2: Research Session - "The Day Of Experiment" (Spring Week 2026)
Goal: Data collection with continuous safeguarding

Step	Operation / Action	Party Involved	Documents & Forms		Timeline / Duration
2.1	Room Setup	Researcher	- Laptops/Tablets, Power supply	FEBRUARY 2026	30 mins pre-session
2.2	Child Assent	Researcher to Pupils	- Child-Participant-Information-Script.pdf - Critical: Verbal script read aloud. Verbal "Yes" required. Children told they can stop anytime.		5–10 mins
2.3	Activity 1: Gameplay	Pupils (Supervised by Teacher)	- Prototype Game: Interactive Gaelic language game. - Note: Teacher present for safeguarding but does not play.		30–40 mins
2.4	Activity 2: Evaluation	Pupils & Teachers	- Pupil-Qualtrics-Questionnaire - Teaching-Staff-Qualtrics-Questionnaire 2 x Format: Digital (Qualtrics) survey		5–10 mins
2.5	Activity 3: Discussion	Researcher & Pupils	- Discussion-Group-Guide.pdf - Format: Conversational group feedback (Field notes taken).		10–15 mins
2.6	Session Close	Researcher	- Distribution of Rewards: 1. Certificate of Participation 2. Small stationery item		Immediate post-session
2.7	Debriefing	Researcher to All	- PG_Debrief_Letter.pdf (Sent home) 2 x - T_Debrief-Letter.pdf (Handed to teacher)		Handout

Phase 3: Post-Study & Data Governance (Week 5-9)
Goal: Anonymization and compliance with withdrawal rights

Step	Operation / Action	Party Involved	Protocol / Rule		Timeline / Deadline
3.1	Pseudonymization	Researcher	- Ensure no names / personal identifiers present, only Participant Codes (e.g., P01, T01). - Securely separate consent forms from data.	MARCH – 3rd APRIL	Immediately after visit
3.2	Withdrawal Window	Parents / Teachers	- Right to Withdraw: Participants may contact researcher to delete data. - Constraint: Must be done within 48 hours of data collection.		48 Hours post-session
3.3	Final Anonymization	Researcher	- After 48h, the link between Code and Name is deleted. - Data is uploaded to University Cloud.		> 48 Hours
3.4	Analysis & Reporting	Researcher	- Analysis of gameplay metrics and questionnaire responses. - Writing of thesis/report.		Up to 4 weeks after

Form Destroy Personal Data Digital Questionnaire Signed Form

Chapter 7 Evaluation and critical appraisal

You should evaluate your own work with respect to your original objectives. You should also critically evaluate your work with respect to related work done by others. You should compare and contrast the project to similar work in the public domain, for example as written about in published papers, or as distributed in software available to you.

Chapter 8 Conclusions

You should summarise your project, emphasising your key achievements and significant drawbacks to your work, and discuss future directions your work could be taken in.

Bibliography

‘Afro-Gaidheil’ (2021) *Trusadh, Series 13*. BBC ALBA. Available at: https://www.bbc.co.uk/programmes/m00109y9.

‘a-nise’ *Faclair na Gàidhlig / The Gaelic Dictionary*. Available at: https://www.faclair.com/ViewEntry.aspx?ID=86992C75F397FF43A461E53D74AA63B7.

Basu, P. (2022) ‘7 Cairns in the Landscape: Migrant Stones and Migrant Stories in Scotland and its Diaspora’, in A. Árnason et al. (eds) *Landscapes Beyond Land*. Berghahn Books, pp. 116-138. Available at: https://doi.org/10.1515/9780857456724-010.

Butgereit, L. (2018) ‘Five Lessons Learned Using Water-Scrum-Fall in South Africa’, *2018 International Conference on Multidisciplinary Research*, 2018, pp. 175-182. Available at: https://doi.org/10.26803/MyRes.2018.14.

Devices for 700,000 children (2021). Scottish Government. Available at: https://www.gov.scot/news/devices-for-700-000-children/.

‘duda’ *LearnGaelic Dictionary*. Bòrd na Gàidhlig (or ‘LearnGaelic.scot’). Available at: https://learngaelic.scot/dictionary/index.jsp?abairt=duda&slang=both&wholeword=false (Accessed: 30 January 2026).

Dunmore, S.S. (2017) ‘Immersion education outcomes and the Gaelic community: identities and language ideologies among Gaelic medium-educated adults in Scotland’, *Journal of Multilingual and Multicultural Development*, 38(8), pp. 726-741. Available at: https://doi.org/10.1080/01434632.2016.1249875.

Education Scotland (2017) *Statutory guidance on Gaelic education*. Available at: https://www.gaidhlig.scot/wp-content/uploads/2017/01/Statutory-Guidance-for-Gaelic-Education.pdf.

Education (Scotland) Act 1872 (no date) *35 & 36 Vict. c. 62*.

‘Ergonomics of human-system interaction - Part 210: Human-centred design for interactive systems’ (2019). Geneva, Switzerland. Available at: https://www.iso.org/standard/77520.html.

Faclair na Gàidhlig / The Gaelic Dictionary (no date) ‘haoidh’. Sabhal Mòr Ostaig. Available at: https://www.faclair.com/ViewEntry.aspx?ID=F499758EDBC7C32081D212BFEC25E067.

Felicya, L. and Lestari, D.P. (2024) ‘Interaction Design of Indonesian Local Language Learning Application Using User-Centered Design’, in *2024 11th International Conference on Advanced Informatics: Concept, Theory and Application (ICAICTA). 2024 11th International*

Conference on Advanced Informatics: Concept, Theory and Application (ICAICTA), Singapore: IEEE, pp. 1-6. Available at: https://doi.org/10.1109/ICAICTA63815.2024.10763276.

Fishman, J.A. (1991) *Reversing Language Shift : theoretical and empirical foundations of assistance to threatened languages*. Clevedon; Philadelphia: Multilingual Matters.

Fraser, B. (1999) 'What are discourse markers?', 31(7), pp. 931-952. Available at: https://doi.org/10.1016/S0378-2166(98)00101-5.

Gaelic Language (Scotland) Act 2005 (2005). Available at: https://www.legislation.gov.uk/asp/2005/7/contents.

Galloway, J. (2011) 'OPPORTUNITIES FOR THE GAELIC VERSIONING OF ELECTRONIC GAMES'.

'giomach, rionnach is ròn, trì seòid a' chuain' (no date) *Faclair na Gàidhlig / Am Faclair Beag (Scottish Gaelic Dictionary)*. Faclair na Gàidhlig (incorporating Dwelly). Available at: https://www.faclair.com/ViewEntry.aspx?ID=115BD9193AA9DC5BF4B7827F66528F8B.

Harbord, C., Lyons, D. and Dempster, E. (2022) 'Playing For Keeps: Digital Games to Preserve Indigenous Languages & Traditions.', in. *Proceedings of DiGRA 2022 Conference: Bringing Worlds Together*. Available at: https://doi.org/10.26503/dl.v2022i1.1384.

Harper, C. and McDougall, K. (2012) 'The Very Recent Fall and Rise of Harris Tweed', *TEXTILE*, 10(1), pp. 78-98. Available at: https://doi.org/10.2752/175183512X13267336595151.

Hunicke, R., LeBlanc, M. and Zubek, R. 'MDA: A Formal Approach to Game Design and Game Research'.

Ishaq, K. *et al.* (2022) 'Serious game design model for language learning in the cultural context', *Education and Information Technologies*, 27(7), pp. 9317-9355. Available at: https://doi.org/10.1007/s10639-022-10999-5.

John W. Creswell and Vicki L Plano Clark (no date) *Designing and conducting mixed methods research*. 2nd edn. Thousand Oaks: Sage Publications.

Johnson, R.B. and Onwuegbuzie, A.J. (2004) 'Mixed Methods Research: A Research Paradigm Whose Time Has Come', *Educational Researcher*, 33(7), pp. 14-26. Available at: https://doi.org/10.3102/0013189X033007014.

Lamb, William (2015) 'Verbal Formulas in Gaelic Traditional Narrative: Some Aspects of Their Form and Function', in *Registers of Communication* . Helsinki, pp. 225-246.

LearnGaelic Dictionary '"cho coltach ri chèile ri dà sgadan."', *LearnGaelic Dictionary*.

LearnGaelic Dictionary 'seadh'. Available at: https://learngaelic.scot/dictionary/index.jsp?abairt=seadh&slang=both&wholeword=false.

MacDonald, Duncan (1882-1954) (1953) *Triùir Mhac Rìgh Èireann agus na Buidsichean (The King of Ireland's Three Sons and the Wizards)*. Available at: https://www.tobarandualchais.co.uk/track/109392?l=en.

MacDonald, T. D. (1926) *Gaelic Proverbs and Proverbial Sayings with English Translations*. Stirling: Eneas Mackay. Available at: https://www3.smo.uhi.ac.uk/gaidhlig/corpus/seanfhaclan/Gaelic_Proverbs_TD_Macdonald.doc.

MacIntyre, Donald, 1899-1964 (1962) *THA AM FIOSRAICHE A' MÌNEACHADH FHACLAN SNA H-ÒRAIN AIGE.* School of Scottish Studies (SoSS). Available at: https://www.tobarandualchais.co.uk/track/101408?l=en%20-%20transcript%20contains%20multiple:%20"Aidh,%20aidh.",%20"Seadh."%20throughout.

McLeod, W. (2014) 'Gaelic in contemporary Scotland: challenges, strategies and contradictions', *europa ethnica*, 71(1-2), pp. 3-12. Available at: https://doi.org/10.24989/0014-2492-2014-12-3.

McLeod, W. (2019) 'The nature of minority languages: insights from Scotland', *Multilingua*, 38(2), pp. 141-154. Available at: https://doi.org/10.1515/multi-2018-0034.

Muhammad, M. *et al.* (2015a) 'Reflections on Researcher Identity and Power: The Impact of Positionality on Community Based Participatory Research (CBPR) Processes and Outcomes', *Critical Sociology*, 41(7-8), pp. 1045-1063. Available at: https://doi.org/10.1177/0896920513516025.

'Nach buidhe dhut' *LearnGaelic - Dictionary*. Available at: https://learngaelic.scot/dictionary/index.jsp?abairt=nach%20buidhe%20dhut&slang=both&wholeword=false.

NicLeòid, S. (2018) '4. Gaelic amongst Schoolchildren: Ideas on Language Change and Linguistic Choices in Gaelic', in MacLeod, M. and Smith-Christmas, C., *Gaelic in Contemporary Scotland*. Edinburgh University Press, pp. 45-61. Available at: https://doi.org/10.1515/9781474420662-011.

Nicolson, Alexander & Macintosh, Rev. Donald (1882) *A Collection of Gaelic Proverbs and Familiar Phrases: Based on Macintosh's Collection*. Edinburgh: Maclachlan and Stewart (Edinburgh); Simpkin, Marshall, & Co. (London). Available at: https://archive.org/details/collectionofgael00maci/page/334/mode/2up.

'och' (no date) *Faclair na Gàidhlig / The Gaelic Dictionary*. Faclair na Gàidhlig (Sabhal Mòr Ostaig). Available at: https://www.faclair.com/ViewEntry.aspx?ID=E01434EFC843E45D5E01BAC9E8040EE9.

O'Hanlon, F., McLeod, W. and Paterson, L. (2010) 'Gaelic-medium Education in Scotland: choice and attainment at the primary and early secondary school stages'.

O'Hanlon, F. and Paterson, L. (2015) '17 Gaelic Education since 1872', in Anderson, R., Freeman, M., and Paterson, L., *The Edinburgh History of Education in Scotland*. Edinburgh University Press, pp. 304-325. Available at: https://doi.org/10.1515/9780748679164-020.

Petri, G. (2019) 'MEEGA+: A Method for the Evaluation of the Quality of Games for Computing Education'.

Phan, M.H., Keebler, J.R. and Chaparro, B.S. (2016) 'The Development and Validation of the Game User Experience Satisfaction Scale (GUESS)', *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 58(8), pp. 1217-1247. Available at: https://doi.org/10.1177/0018720816669646.

Pruitt, J. and Grudin, J. (2003) 'Personas: practice and theory', in *Proceedings of the 2003 conference on Designing for user experiences*. *DUX03: Designing the User Experience*, San Francisco California: ACM, pp. 1-15. Available at: https://doi.org/10.1145/997078.997089.

Rodríguez, I. *et al.* (2020) 'Evaluating the effect of gamification on the deployment of digital cultural probes for children', *International Journal of Human-Computer Studies*, 137, p. 102395. Available at: https://doi.org/10.1016/j.ijhcs.2020.102395.

Schwaber, K. (1997) 'SCRUM Development Process', in J. Sutherland et al. (eds) *Business Object Design and Implementation*. London: Springer London, pp. 117-134. Available at: https://doi.org/10.1007/978-1-4471-0947-1_11.

'Scottish Languages Act 2025'. Available at: https://www.legislation.gov.uk/asp/2025/10/contents.

Sommerville, Ian (2016) *Software Engineering*. Tenth edition, Global edition. Pearson Education.

SpeakGaelic (2024) 'Sin thu fhèin! (Phrases - A' dol a chèilidh)'. BBC / MG ALBA.

Stornoway Gazette (2021) 'Stornoway Primary School boy wins Gaelic award four years after arriving from Syria', April. Available at: https://www.stornowaygazette.co.uk/education/stornoway-primary-school-boy-wins-gaelic-award-four-years-after-arriving-from-syria-3213796.

Thomas, D.R. (2003) 'A general inductive approach for qualitative data analysis'.

Timmermans, S. and Tavory, I. (2012) 'Theory Construction in Qualitative Research: From Grounded Theory to Abductive Analysis', *Sociological Theory*, 30(3), pp. 167-186. Available at: https://doi.org/10.1177/0735275112457914.

Veresov, N. and Veraksa, N. (2023) 'Digital games and digital play in early childhood: a cultural-historical approach', *Early Years*, 43(4-5), pp. 1089-1101. Available at: https://doi.org/10.1080/09575146.2022.2056880.

Wang, W. (2023) *The Structure of Game Design*. Cham, Switzerland: Springer Nature Switzerland AG (International Series on Computer, Entertainment and Media Technology).

Zhang, X. *et al.* (2025) ‘How cultural elements shape game evaluations: The role of cultural authenticity and perceived effort’, *Computers in Human Behavior*, 162, p. 108452. Available at: https://doi.org/10.1016/j.chb.2024.108452.

Appendix

- TBD **Appendix**: user manual Instructions on installing, executing and using the system where appropriate.

Stakeholder Meeting 1 Field Notes: 16/01/2026

Meeting with myself the researcher (R) headteacher (HT) only. This was to describe the schools technical setup to ensure the game was developed for the right device type.
[R] Thank you for joining me, is it ok if I take notes of my questions and your responses during our conversation? [HT] Yes, not a problem. [R] Question: How comfortable are staff and children using technology in the P6/7 cohort, and what setup do they have? [HT]"Yes, the pupils and staff in the class are comfortable with tech and use devices regularly. They have a selection of either a Chromebook or iPad. I'd say the iPads are faster, but we tend to prefer using the Chromebooks as they have keyboards and are touchscreen too." [R] "Do those devices listed have facilities for sound and screen recording?" [HT] "I'm not sure about screen recording…but the chromebooks definitely have audio jacks" [R] Question: Would you be interested in looking at prototypes and giving feedback? [HT] "Absolutely, just myself? Other staff? Children?" [R] Yourself and the member of staff that teaches the class I will conduct the research session would be great. [R] Do you have any questions for me at this stage? [HT] No, but I'm looking forward to seeing the games soon. How are you getting on with it all? [R] Thank you, the game content itself has been designed and implementation is underway. However, your thoughts on the games early on are highly valuable. Shall we agree on a date to show you what I have at this stage and we can discuss any improvements? *Compared diaries and settled on the afternoon 21/01/2026 *

School of Computer Science Ethics Committee

19 December 2025

Dear Luke Macleod

Thank you for submitting an Ethics Application Form application for review by the Computer Science ethics committee. The committee reviewed this application and accompanying documents on 19 December 2025. The outcome of this review is given below:

Project Title User-Centred Design of Games for Gaelic Digital Normalisation
Researcher(s) Luke Macleod
Supervisor(s) Dr Miriam Sturdee
Application Ref 0864 - CS-0864-1388-2025
Decision Date 19 December 2025 **Decision Expiry Date** 19 December 2030
Review Outcome Favourable opinion
Specific Conditions *(optional)* None
Ethics committee comments *(optional)* None

The following supporting documents are acknowledged:

Document Type	File Name	Date	Version
Recruitment Document	letter-to-parentguardian.pdf	30/11/2025	1.2
Recruitment Document	letter-to-school-in-principle.pdf	30/11/2025	1.2
Recruitment Document	letter-to-school-commencing.pdf	30/11/2025	1.2
PIS Document	PG_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2
PIS Document	Child Participant Information Script.pdf	30/11/2025	1.0
PIS Document	T_PIS_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2
Participant Consent Document	Child Participant Information Script.pdf	30/11/2025	1
Participant Consent Document	T_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2
Participant Consent Document	PG_Consent form_[NOV 2025]_[V1.2]_[USER-CENTRED DESIGN OF GAMES FOR GAELIC DIGITAL NORMALISATION].pdf	30/11/2025	1.2
Evidence of Permission Document	school-participation-agreement-email.pdf		
Evidence of Permission Document	CNES-e-mail-confirmation.pdf		
Default	Discussion Group Guide.pdf	30/11/2025	1.0
Debrief Document or Protocol	T_Debrief-Letter.pdf	30/11/2025	1.0
Debrief Document or Protocol	PG_Debrief_Letter.pdf	30/11/2025	1.0
Application Supporting Document	PVG Guidance.png		
Application Supporting Document	Pupil Qualtrics Questionnaire Questions.pdf	30/11/2025	1.0
Application Supporting Document	Teaching Staff Qualtrics Questionnaire Questions.pdf	30/11/2025	1.0

Document Type	File Name	Date	Version
Application Supporting Document	Ethics-Phases-of-Significance.pdf	01/12/2025	1.0

Favourable opinions

A favourable opinion is conditional upon any conditions set by the committee, if any, as described in the 'Specific conditions' section above.

A favourable opinion is valid for 5 years from the decision date, see the expiry date above.

If you wish for this opinion to apply to any subsequent changes made to the project, you must first submit an amendment request to the ethics committee, using the University's ethics amendment application form. Changes made to the research without the submission of such a request will invalidate this favourable opinion.

Ethics opinions must be renewed every 5 years by submission of a new application. If only a short extension is required, for example to finish writing up, you can request a discretionary extension of up to 6 months from the ethics committee.

You must report any serious adverse events, or significant changes not covered by this approval, related to this study immediately to the ethics committee.

A favourable opinion is given on the condition that:

- you abide by any specific conditions set by the ethics committee
- you conduct your research in line with:
  - the details provided in your ethical application.
  - relevant University policies and procedures, including the Principles of Good Research Conduct.
  - the conditions of any funding associated with your work.
  - any local legal or ethical requirements.
- all applicable approvals, permissions, or documents are obtained before research commences.

A favourable opinion by a University ethics committee does not confer any kind of approval for the research, be it governance, legal or otherwise. However, a favourable opinion is necessary for the research to proceed.

You should retain this approval letter with your study paperwork.

Yours sincerely,

Computer Science